



# Evolutionary genetics

The Iran Research Institute of Philosophy  
24 August 2019

Dr. Ata Kalirad  
School of Biological Sciences (IPM)

# In the beginning ...

On the Tendency of Species to form Varieties; and on the Perpetuation of Varieties and Species by Natural Means of Selection. By CHARLES DARWIN, Esq., F.R.S., F.L.S., & F.G.S., and ALFRED WALLACE, Esq. Communicated by Sir CHARLES LYELL, F.R.S., F.L.S., and J. D. HOOKER, Esq., M.D., V.P.R.S., F.L.S., &c.

[Read July 1st, 1858.]

London, June 30th, 1858.

MY DEAR SIR,—The accompanying papers, which we have the honour of communicating to the Linnean Society, and which all relate to the same subject, viz. the Laws which affect the Production of Varieties, Races, and Species, contain the results of the investigations of two indefatigable naturalists, Mr. Charles Darwin and Mr. Alfred Wallace.

These gentlemen having, independently and unknown to one another, conceived the same very ingenious theory to account for the appearance and perpetuation of varieties and of specific forms on our planet, may both fairly claim the merit of being original thinkers in this important line of inquiry; but neither of them having published his views, though Mr. Darwin has for many years past been repeatedly urged by us to do so, and both authors having now unreservedly placed their papers in our hands, we think it would best promote the interests of science that a selection from them should be laid before the Linnean Society.

# In the beginning ...

ANNALS OF EUGENICS  
A JOURNAL FOR THE SCIENTIFIC STUDY  
OF  
RACIAL PROBLEMS

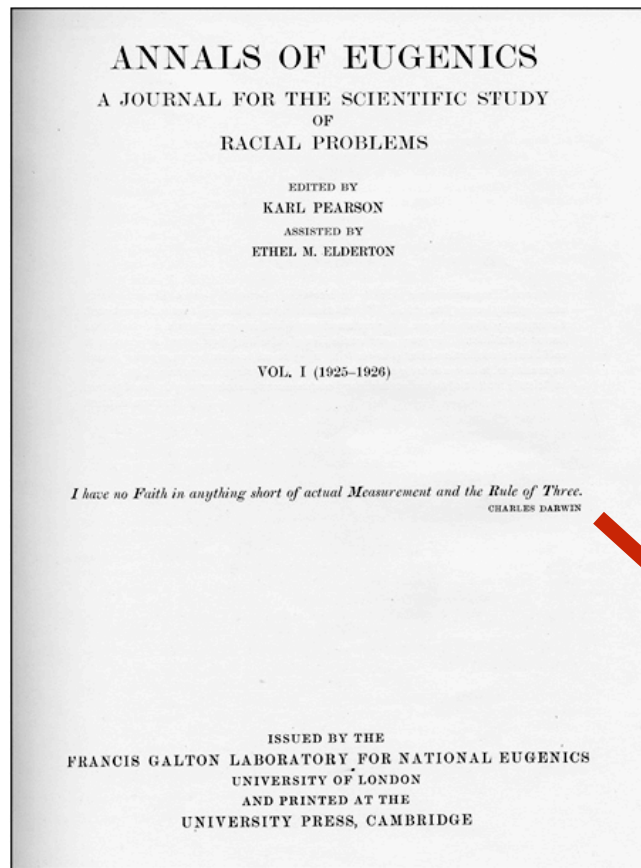
EDITED BY  
KARL PEARSON  
ASSISTED BY  
ETHEL M. ELDERTON

VOL. I (1925-1926)

*I have no Faith in anything short of actual Measurement and the Rule of Three.*  
CHARLES DARWIN

ISSUED BY THE  
FRANCIS GALTON LABORATORY FOR NATIONAL EUGENICS  
UNIVERSITY OF LONDON  
AND PRINTED AT THE  
UNIVERSITY PRESS, CAMBRIDGE

# In the beginning ...



I have no faith in anything short of actual measurement  
and the Rule of Three.

- Charles Darwin

# To measure inheritance

1



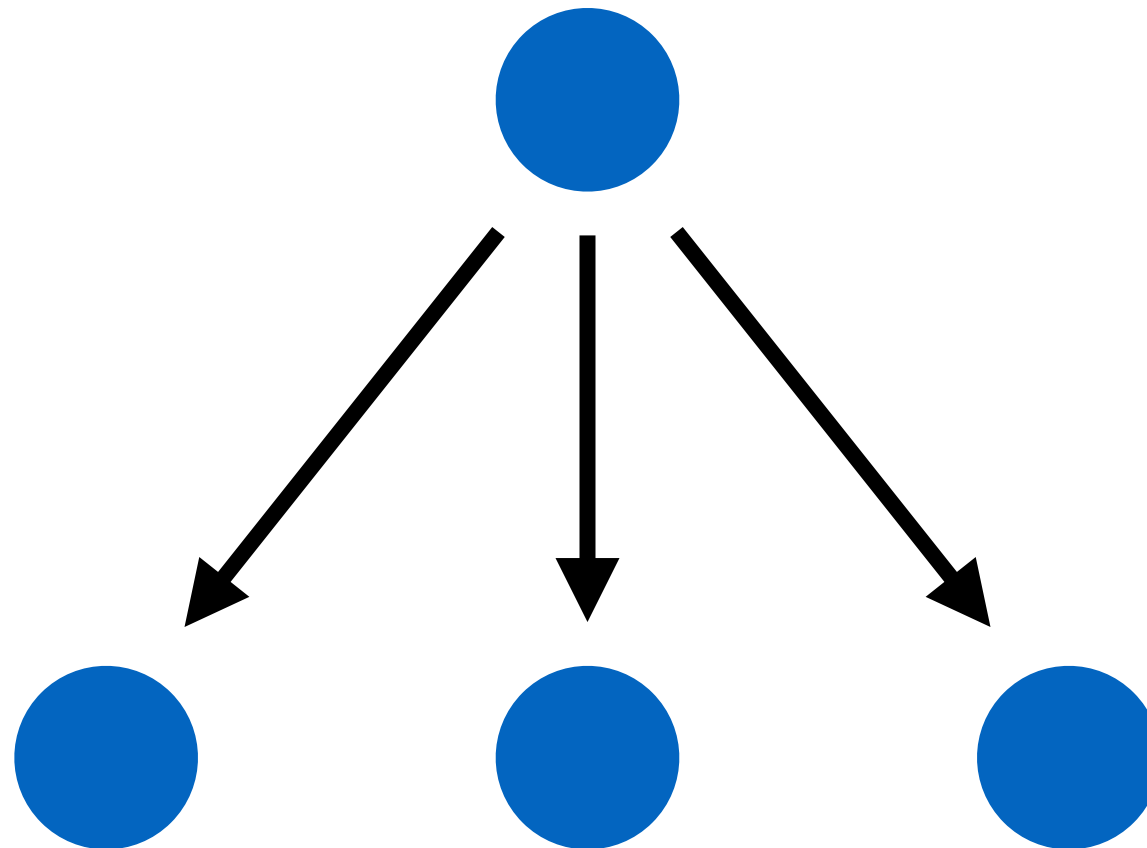
**Sir Francis Galton (1822-1911)**

# To measure inheritance

## Darwin's model of variation

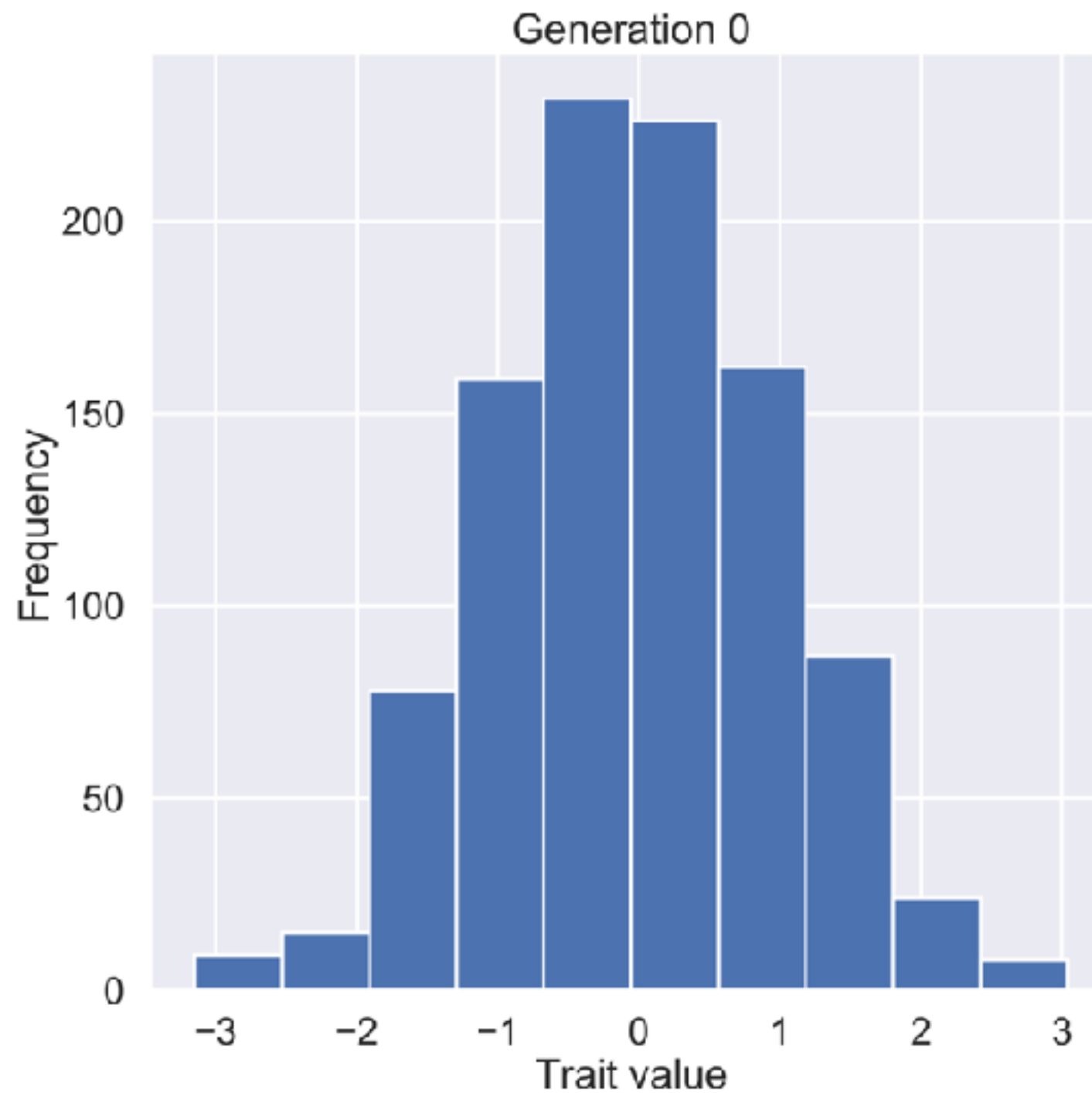
Gen 0

Gen 1



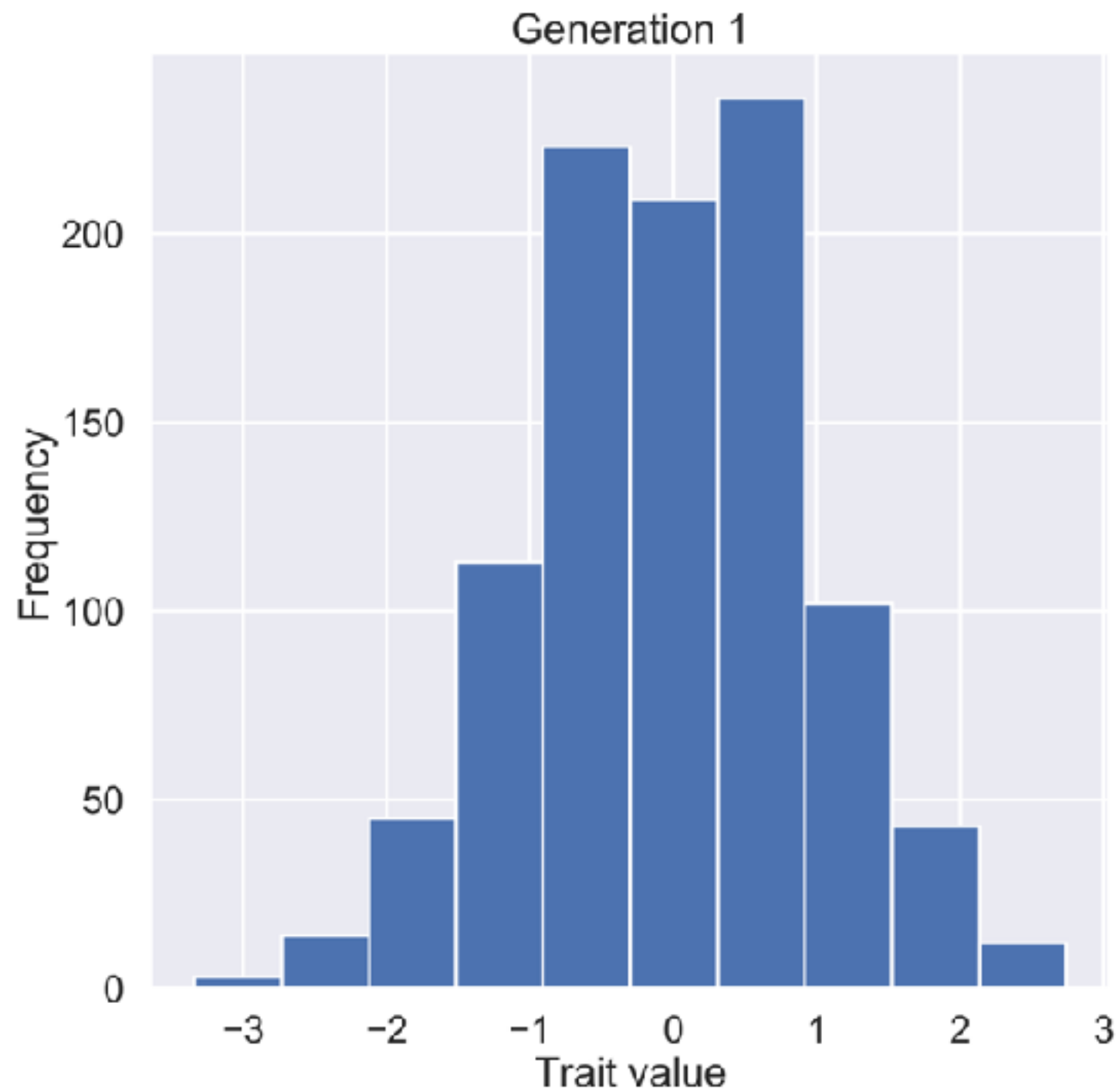
# To measure inheritance

## Galton's observation



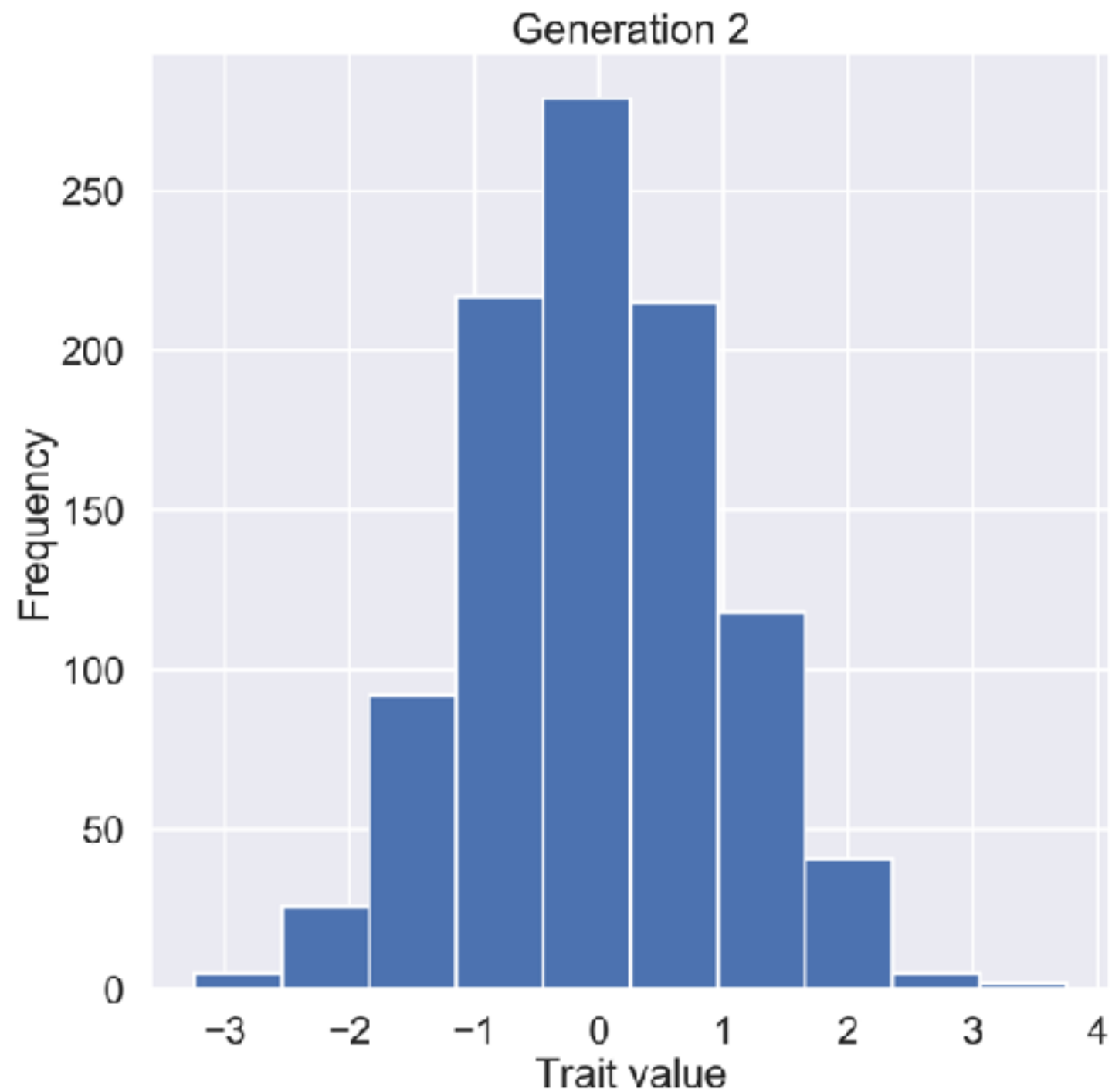
# To measure inheritance

## Galton's observation

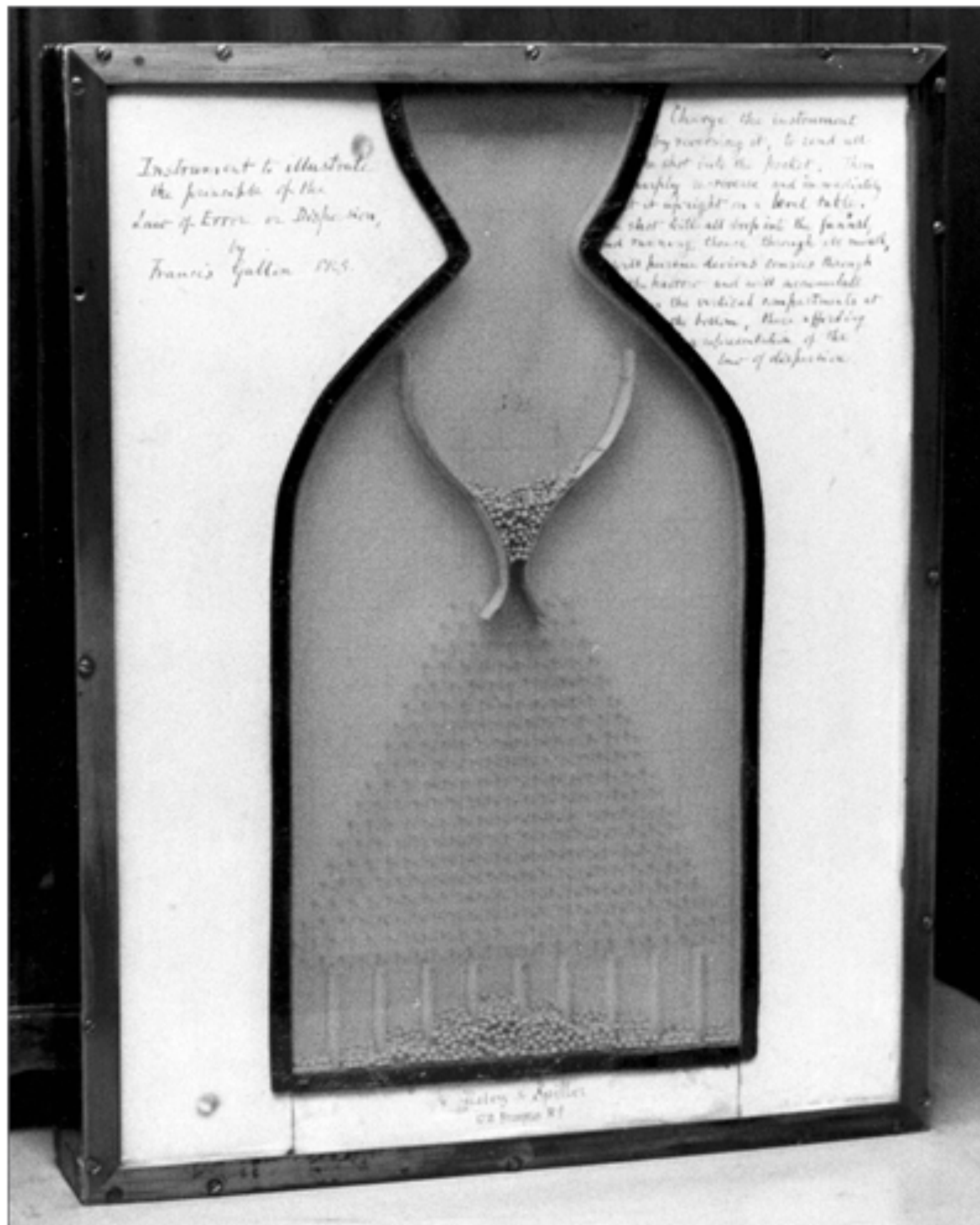


# To measure inheritance

## Galton's observation

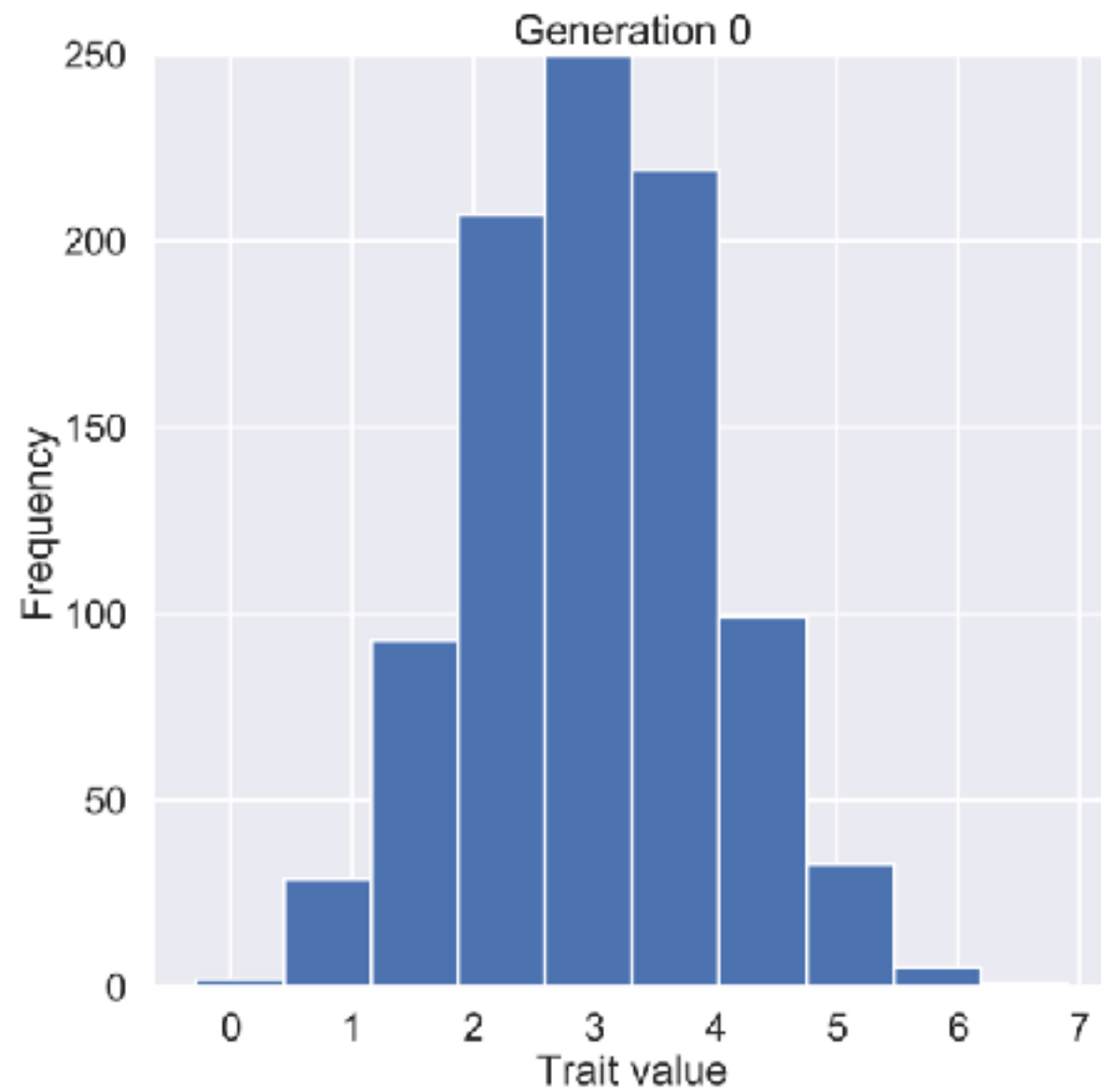


# To measure inheritance

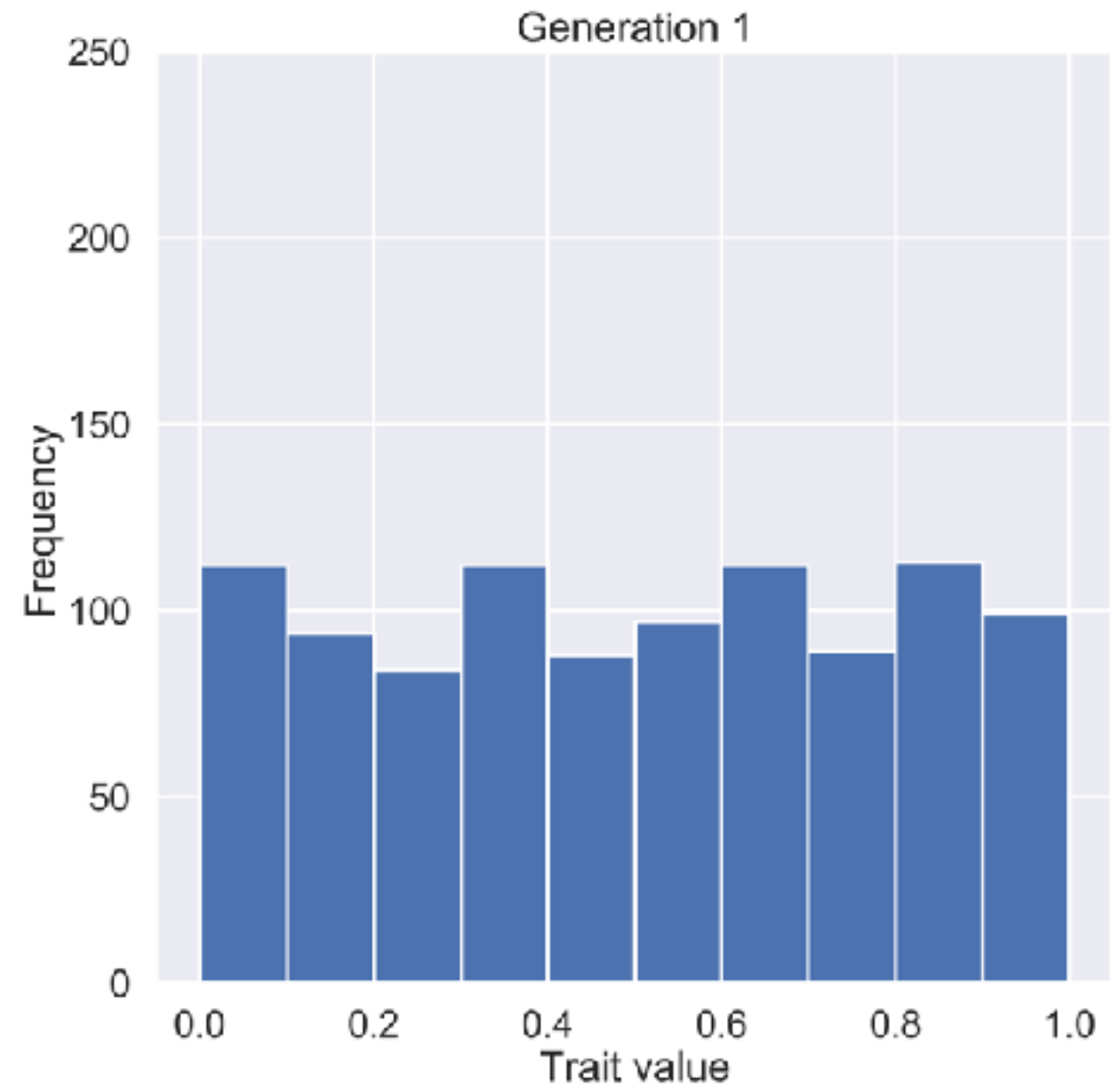
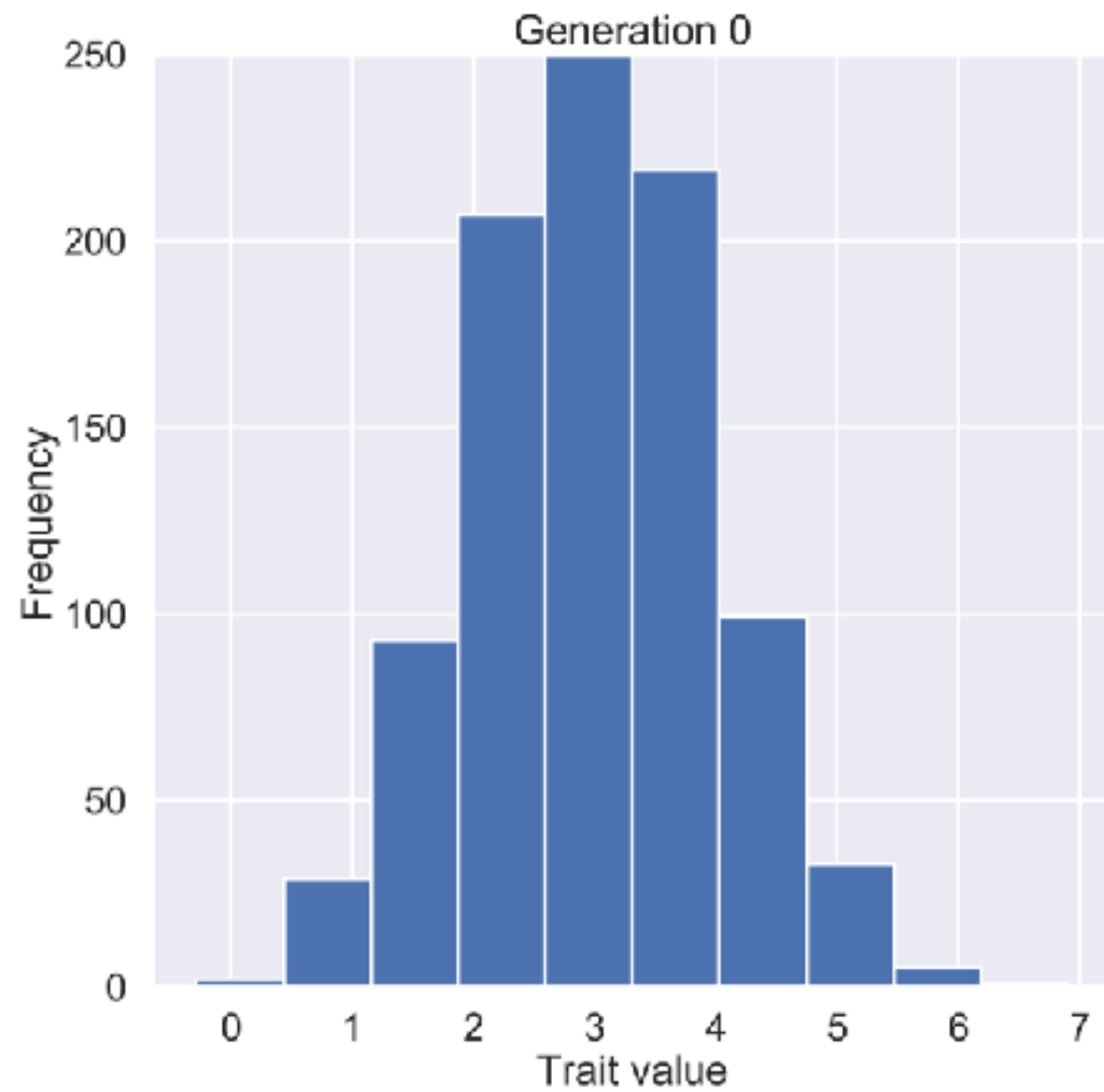


The original quincunx, constructed in 1873 for use in an 1874 public lecture.

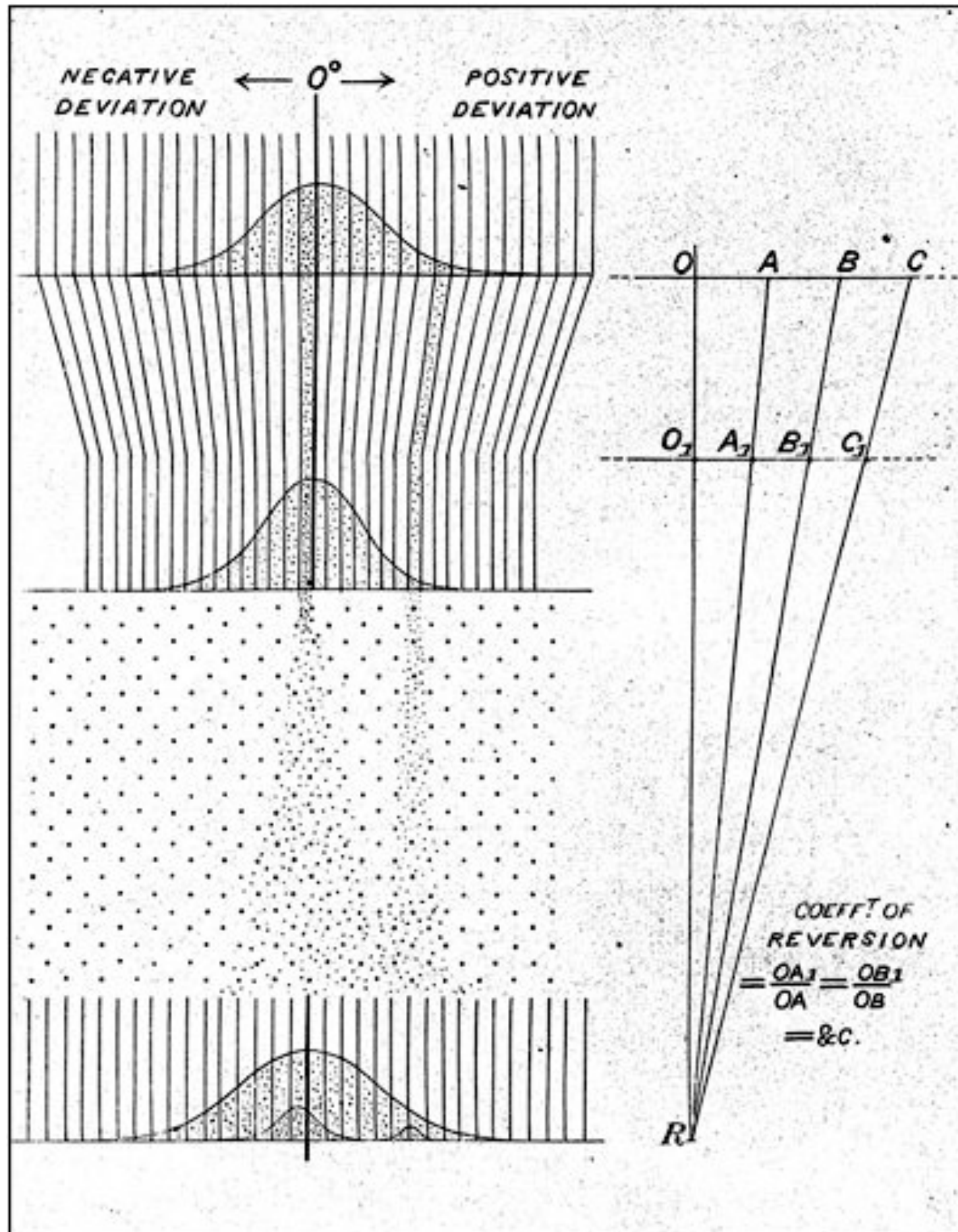
# To measure inheritance



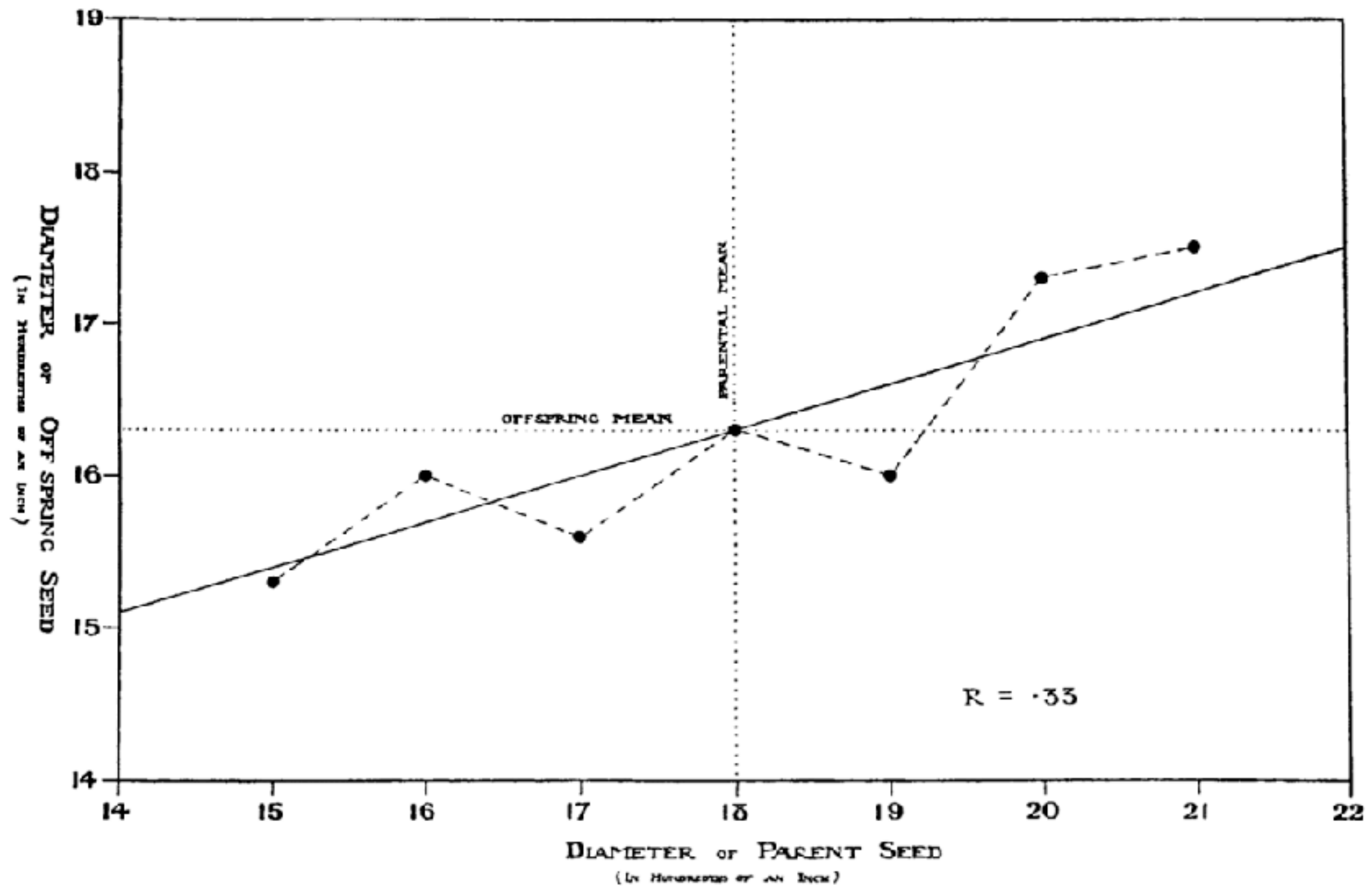
# To measure inheritance



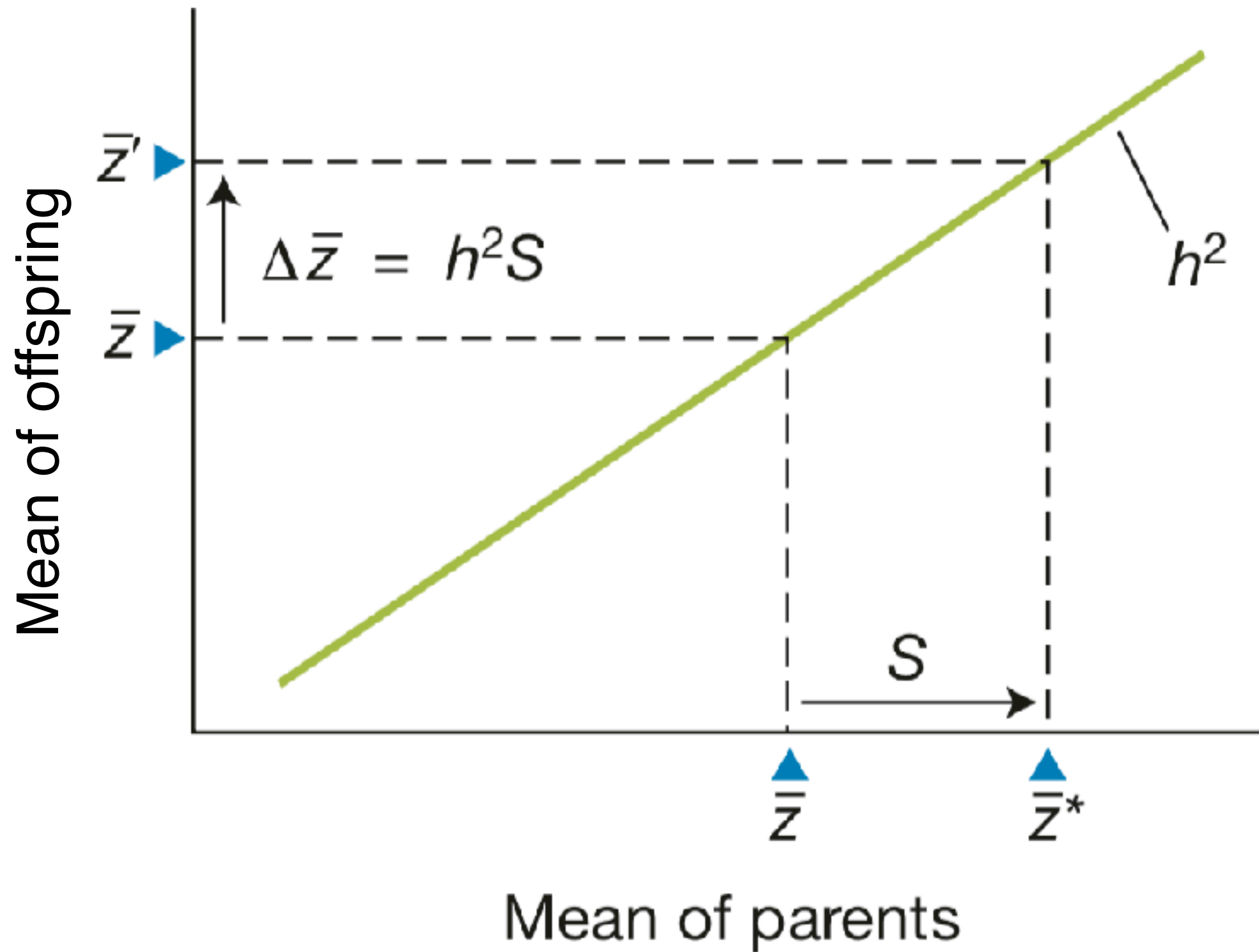
# To measure inheritance



# To measure inheritance



# To measure inheritance



# Is variation random?

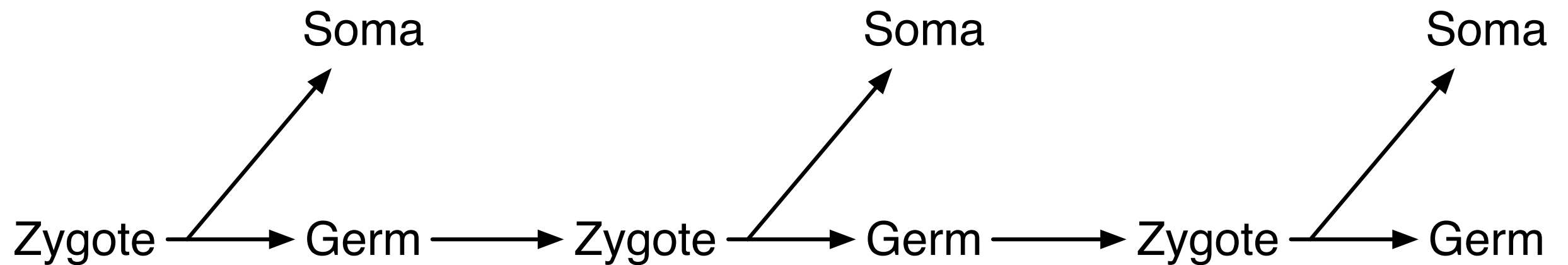
# 2



**Joshua (1925 - 2008) and Esther Lederberg (1922 - 2006)**

# Inheritance of variation

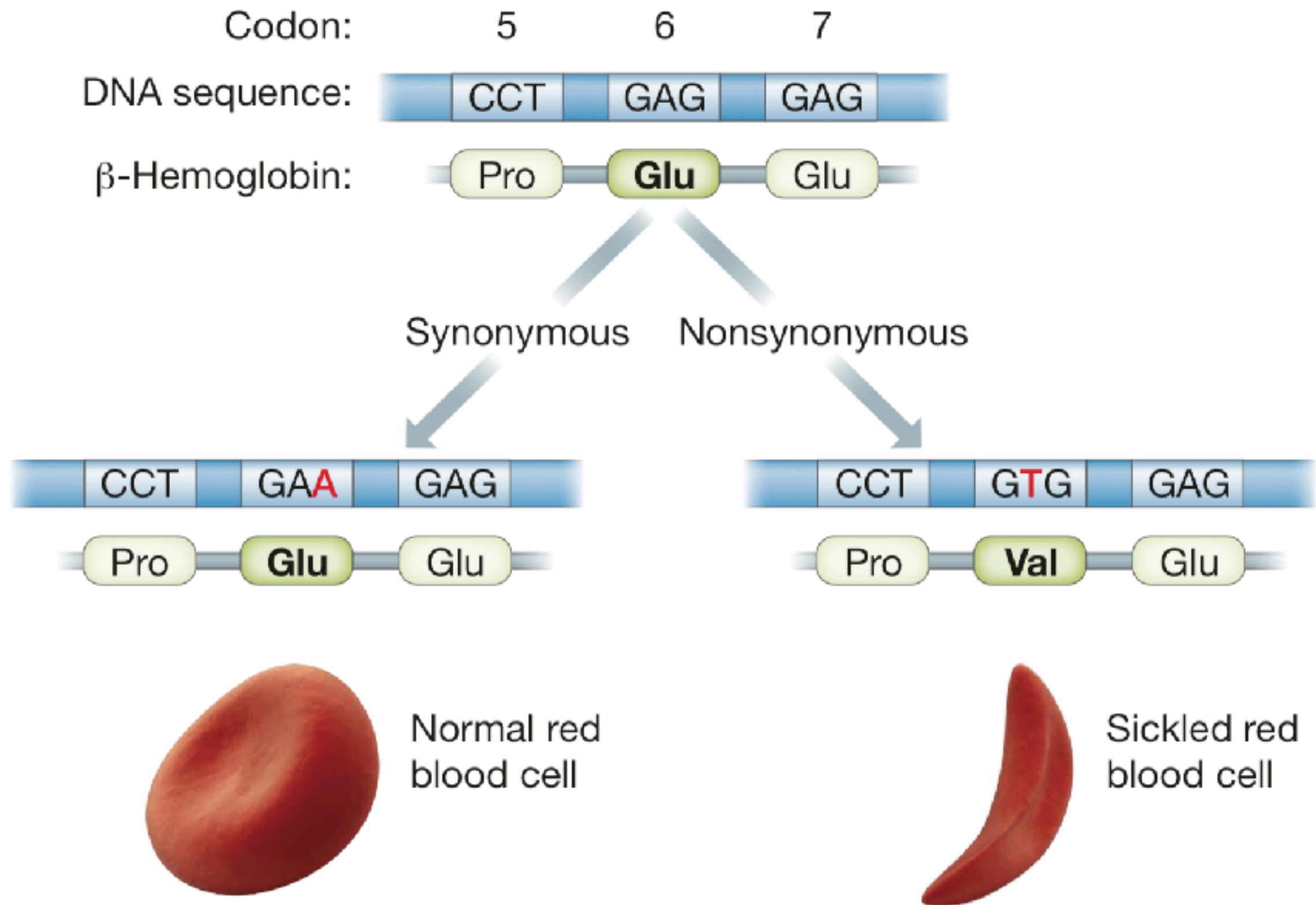
## Mutation: Germ line vs Soma



# The redundancy of the genetic code

|              |   | Second letter                                                             |                                    |                                                                           |                                                                    |
|--------------|---|---------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------|
|              |   | T                                                                         | C                                  | A                                                                         | G                                                                  |
| First letter | T | TTT Phenyl-<br>TTC alanine<br><br>TTA Leucine<br>TTG                      | TCT<br>TCC Serine<br>TCA<br>TCG    | TAT Tyrosine<br>TAC<br><br><b>TAA</b> Stop codon<br><b>TAG</b> Stop codon | TGT Cysteine<br>TGC<br><br><b>TGA</b> Stop codon<br>TGG Tryptophan |
|              | C | CTT<br>CTC Leucine<br>CTA<br>CTG                                          | CCT<br>CCC Proline<br>CCA<br>CCG   | CAT Histidine<br>CAC<br><br>CAA Glutamine<br>CAG                          | CGT<br>CGC Arginine<br>CGA<br>CGG                                  |
|              | A | ATT<br>ATC Isoleucine<br>ATA<br><br><b>ATG</b> Methionine;<br>start codon | ACT<br>ACC Threonine<br>ACA<br>ACG | AAT Asparagine<br>AAC<br><br>AAA Lysine<br>AAG                            | AGT Serine<br>AGC<br><br>AGA Arginine<br>AGG                       |
|              | G | GTT<br>GTC Valine<br>GTA<br>GTG                                           | GCT<br>GCC Alanine<br>GCA<br>GCG   | GAT Aspartic<br>GAC acid<br><br>GAA Glutamic<br>GAG acid                  | GGT<br>GGC Glycine<br>GGA<br>GGG                                   |

# The effect of mutations

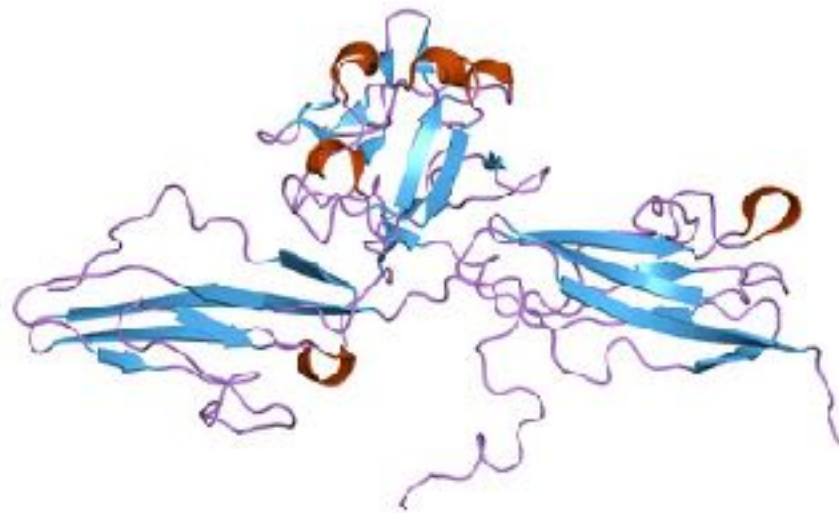


# Inheritance of variation

## Mutation: Effects

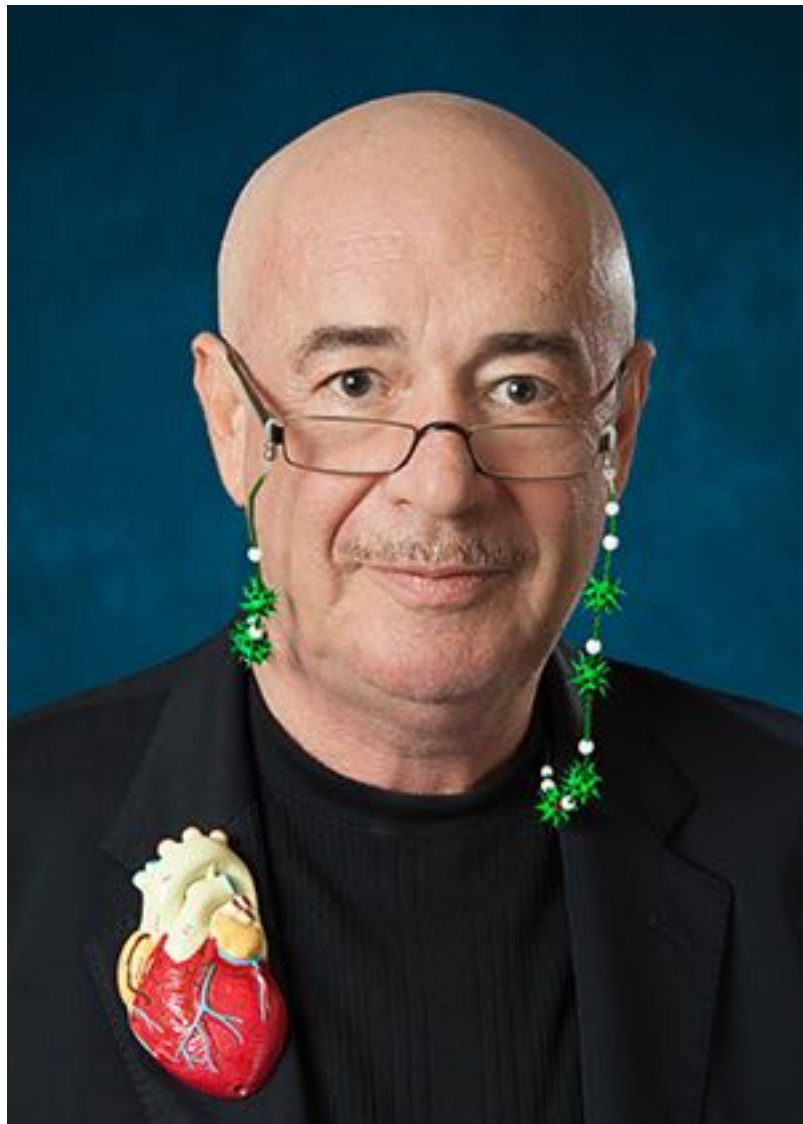
Pleiotropy : Fibroblast growth factor receptor 3

gtcgcgggca gctggcgccg ... aacctggaaa aaaaaaaaaa aaaa



# Inheritance of variation

## Mutation: Are they random?

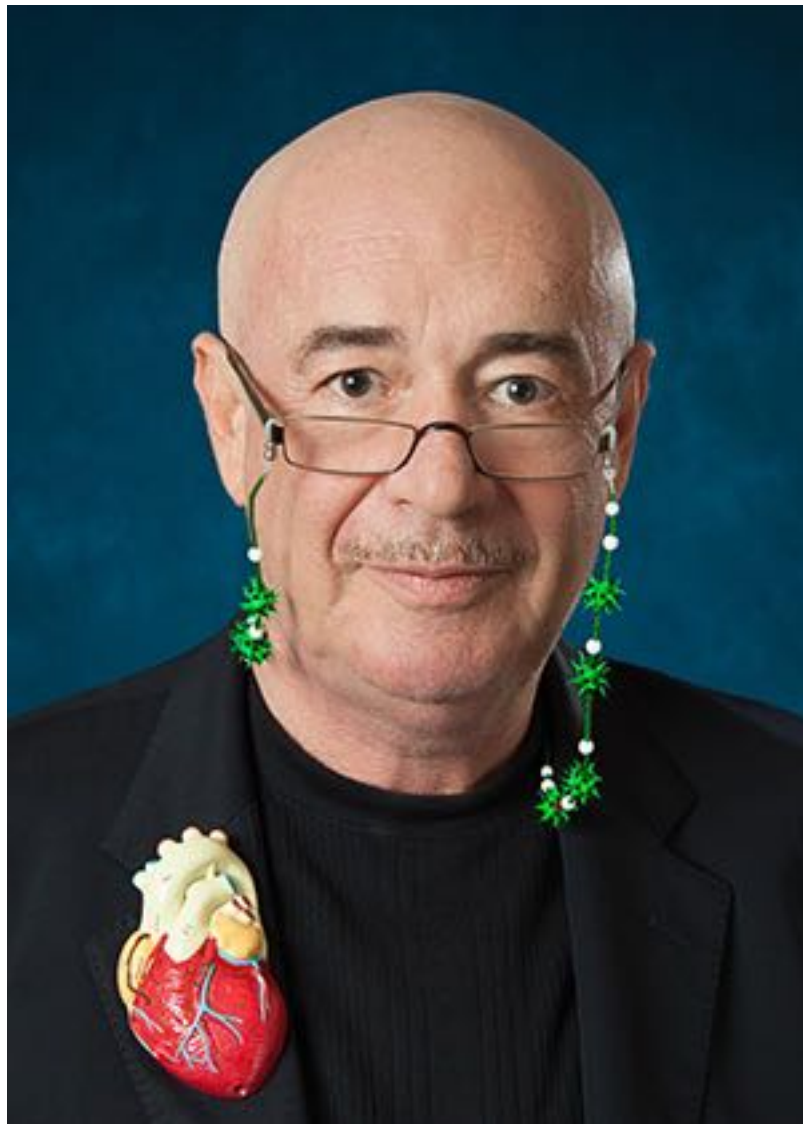


**Dan Graur, UH**

Mutations are commonly said to occur “randomly.” as a matter of fact, the phrase “**mutations are random**” appears in the literature thousands of times. However, empirical studies show unambiguously that mutations do not occur at random with respect to **genomic location** (there are hotspots and coldspots of mutation), **mutation type** (e.g., in most organisms transitions occur more frequently than transversions), or **fitness effects** (nonadvantageous mutations occur more frequently than advantageous ones).

# Inheritance of variation

## Mutation: Are they random?



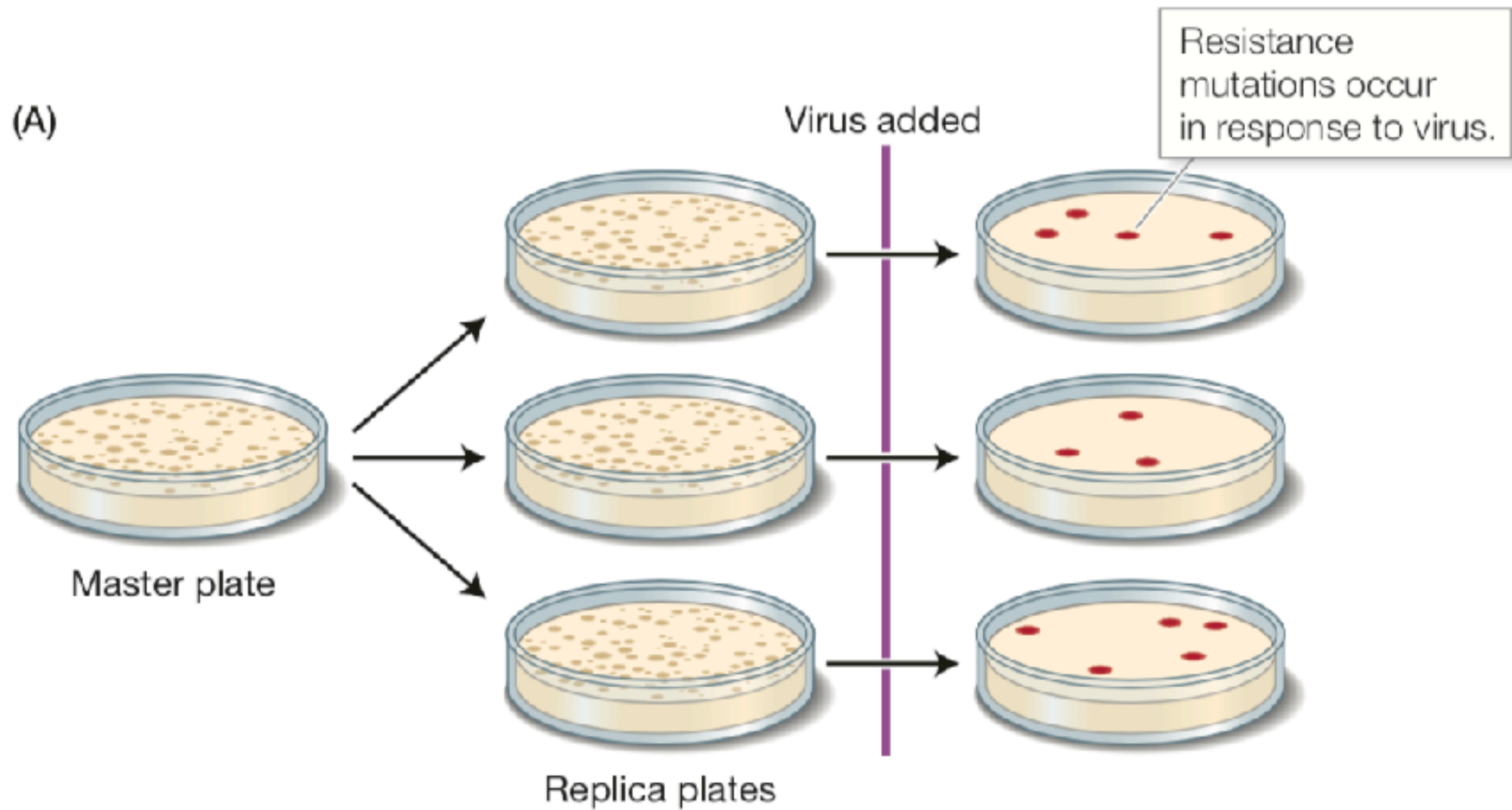
**Dan Graur, UH**

Mutations are said to be random in one respect: “A mutation is expected to occur with the same frequency under conditions in which it confers an advantage on the organism carrying it, confers no advantage, or is deleterious.”

Sadly, this definition is misleading researchers, students, and the general public. This definition has nothing to do with randomness. It would be much clearer to state that **mutations are indifferent** to their effect on fitness.

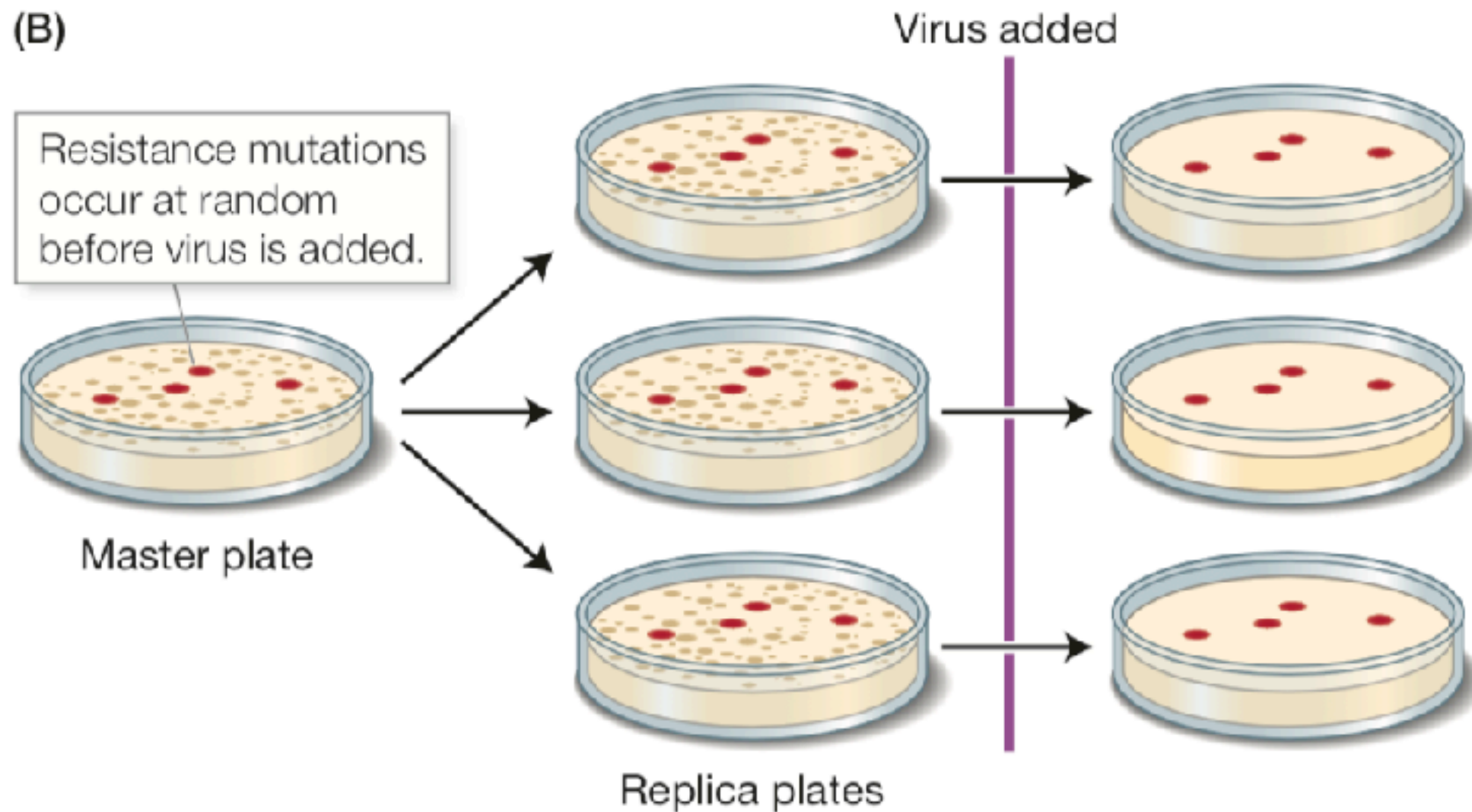
# Inheritance of variation

Mutation: Are they random?



# Inheritance of variation

## Mutation: Are they random?



# What is a non-evolving population?

3



G. H. Hardy (1877 – 1947)

# Inheritance of variation



# Inheritance of variation

JULY 10, 1908

*SCIENCE*

## *DISCUSSION AND CORRESPONDENCE*

### Mendelian Proportions in a Mixed Population

To The Editor of Science: I am reluctant to intrude in a discussion concerning matters of which I have no expert knowledge, and I should have expected the very simple point which I wish to make to have been familiar to biologists.

# Inheritance of variation

## Genotypes

- ▶  $A_1A_1$       $x$
- ▶  $A_1A_2$       $y$
- ▶  $A_2A_2$       $z$
- ▶  $x + y + z = 1$

## Alleles

- ▶  $A_1$       $p$
- ▶  $A_2$       $q$
- ▶  $p + q = 1$
- ▶  $p = x + y/2$

# Inheritance of variation

## Random mating

| Mating                 | Probability |
|------------------------|-------------|
| $A_1A_1 \times A_1A_1$ | $x^2$       |
| $A_1A_1 \times A_1A_2$ | $2xy$       |
| $A_1A_1 \times A_2A_2$ | $2xz$       |
| $A_1A_2 \times A_1A_2$ | $y^2$       |
| $A_1A_2 \times A_2A_2$ | $2yz$       |
| $A_2A_2 \times A_2A_2$ | $z^2$       |

# Inheritance of variation

## Outcome mating

| Mating                 | $A_1A_1$ | $A_1A_2$ | $A_2A_2$ |
|------------------------|----------|----------|----------|
| $A_1A_1 \times A_1A_1$ | 1        | 0        | 0        |
| $A_1A_1 \times A_1A_2$ |          |          |          |
| $A_1A_1 \times A_2A_2$ |          |          |          |
| $A_1A_2 \times A_1A_2$ |          |          |          |
| $A_1A_2 \times A_2A_2$ |          |          |          |
| $A_2A_2 \times A_2A_2$ |          |          |          |

# Inheritance of variation

## Outcome mating

| Mating                 | $A_1A_1$ | $A_1A_2$ | $A_2A_2$ |
|------------------------|----------|----------|----------|
| $A_1A_1 \times A_1A_1$ | 1        | 0        | 0        |
| $A_1A_1 \times A_1A_2$ | 1/2      | 1/2      | 0        |
| $A_1A_1 \times A_2A_2$ | 0        | 1        | 0        |
| $A_1A_2 \times A_1A_2$ | 1/4      | 1/2      | 1/4      |
| $A_1A_2 \times A_2A_2$ | 0        | 1/2      | 1/2      |
| $A_2A_2 \times A_2A_2$ | 0        | 0        | 1        |

# Inheritance of variation

## Outcome mating

| Mating                 | Probability | $A_1A_1$ | $A_1A_2$ | $A_2A_2$ |
|------------------------|-------------|----------|----------|----------|
| $A_1A_1 \times A_1A_1$ | $x^2$       | 1        | 0        | 0        |
| $A_1A_1 \times A_1A_2$ | $2xy$       | 1/2      | 1/2      | 0        |
| $A_1A_1 \times A_2A_2$ | $2xz$       | 0        | 1        | 0        |
| $A_1A_2 \times A_1A_2$ | $y^2$       | 1/4      | 1/2      | 1/4      |
| $A_1A_2 \times A_2A_2$ | $2yz$       | 0        | 1/2      | 1/2      |
| $A_2A_2 \times A_2A_2$ | $z^2$       | 0        | 0        | 1        |

# Inheritance of variation

## Outcome mating

| Mating                 | $A_1A_1$ | $A_1A_2$ | $A_2A_2$ |
|------------------------|----------|----------|----------|
| $A_1A_1 \times A_1A_1$ | $x^2$    | 0        | 0        |
| $A_1A_1 \times A_1A_2$ | $xy$     | $xy$     | 0        |
| $A_1A_1 \times A_2A_2$ | 0        | $2xz$    | 0        |
| $A_1A_2 \times A_1A_2$ | $y^2/4$  | $y^2/2$  | $y^2/4$  |
| $A_1A_2 \times A_2A_2$ | 0        | $yz$     | $yz$     |
| $A_2A_2 \times A_2A_2$ | 0        | 0        | $z^2$    |

# Inheritance of variation

## Next generation

### Genotypes

- ▶  $A_1 A_1$        $x' = p^2$
- ▶  $A_1 A_2$        $y' = 2pq$
- ▶  $A_2 A_2$        $z' = q^2$

### Alleles

- ▶  $A_1$        $p' = p$
- ▶  $A_2$        $q' = q$

# Inheritance of variation

## Hardy-Weinberg equilibrium (Our Null model)

### Genotypes

- ▶  $A_1 A_1$        $x' = p^2$
- ▶  $A_1 A_2$        $y' = 2pq$
- ▶  $A_2 A_2$        $z' = q^2$

### Alleles

- ▶  $A_1$        $p' = p$
- ▶  $A_2$        $q' = q$

The key conditions for the Hardy-Weinberg equilibrium are:

- An infinite population size
- No natural selection
- No mutation
- No movement between populations
- Random mating

# Inheritance of variation

## Natural Selection

**If:**

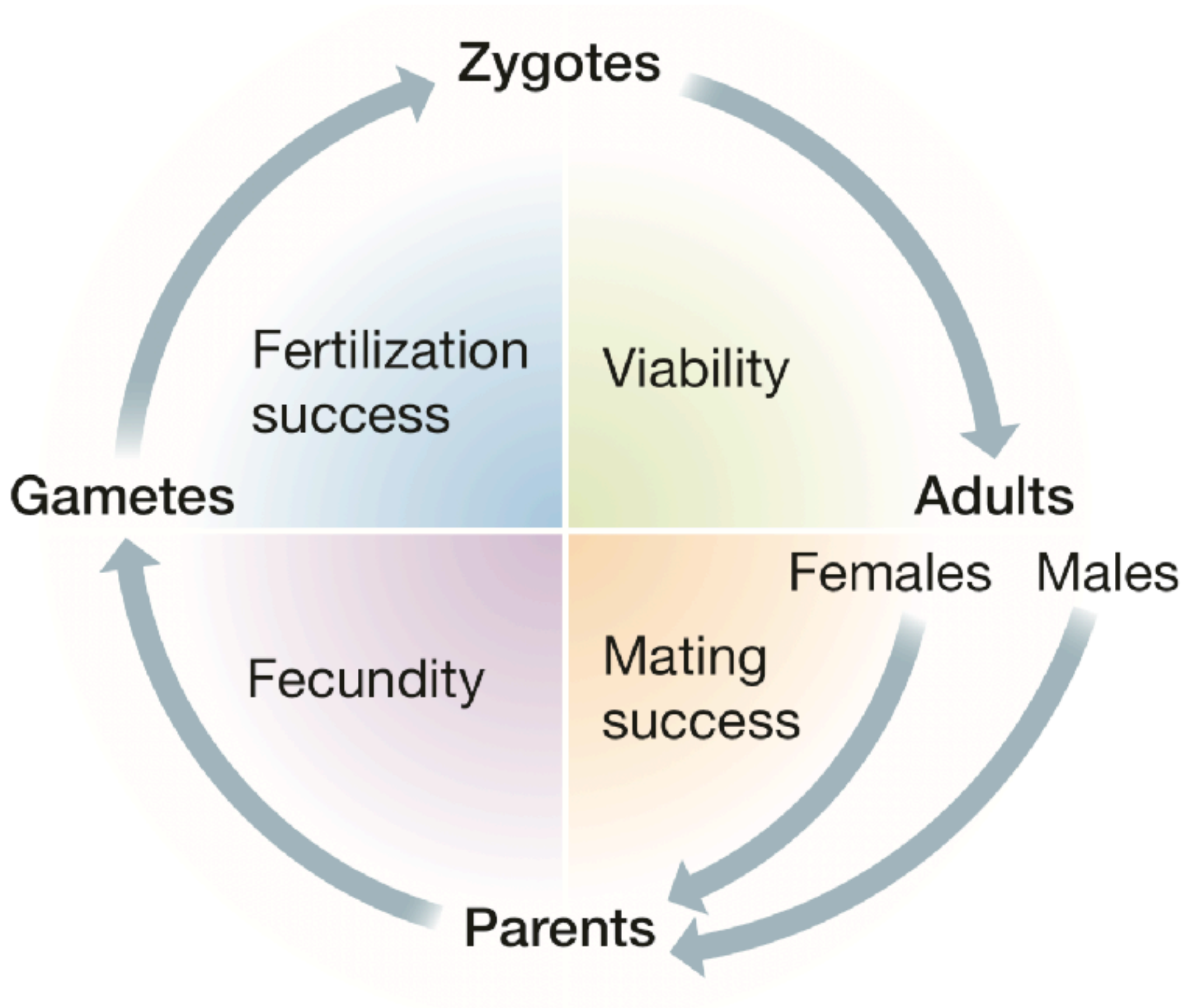
- there is a correlation between a phenotypic trait and the number of offspring that individuals leave to the next generation, and
- there is a correlation between the phenotype of a trait in parents and in their offspring,

**Then:**

- the trait will evolve.

# Inheritance of variation

Fitness



# Inheritance of variation

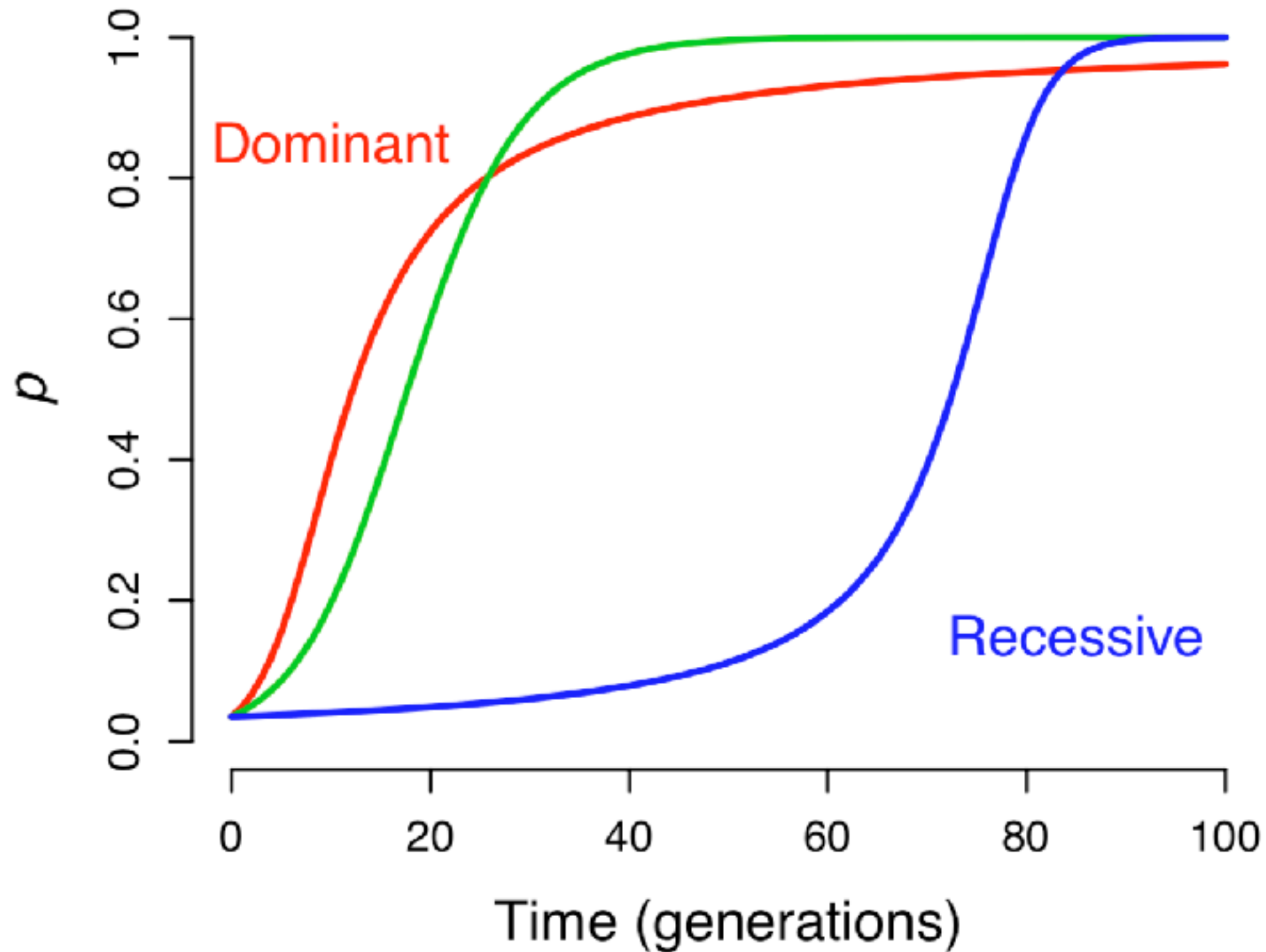
## Adding selection

|                              | $A_1A_1$          | $A_1A_2$          | $A_2A_2$          |
|------------------------------|-------------------|-------------------|-------------------|
| Frequency (before selection) | $x$               | $y$               | $z$               |
| Fitness                      | $w_{11}$          | $w_{12}$          | $w_{22}$          |
| Frequency (after selection)  | $xw_{11}/\bar{w}$ | $yw_{12}/\bar{w}$ | $zw_{22}/\bar{w}$ |

Mean fitness:  $\bar{w} = xw_{11} + yw_{12} + zw_{22}$

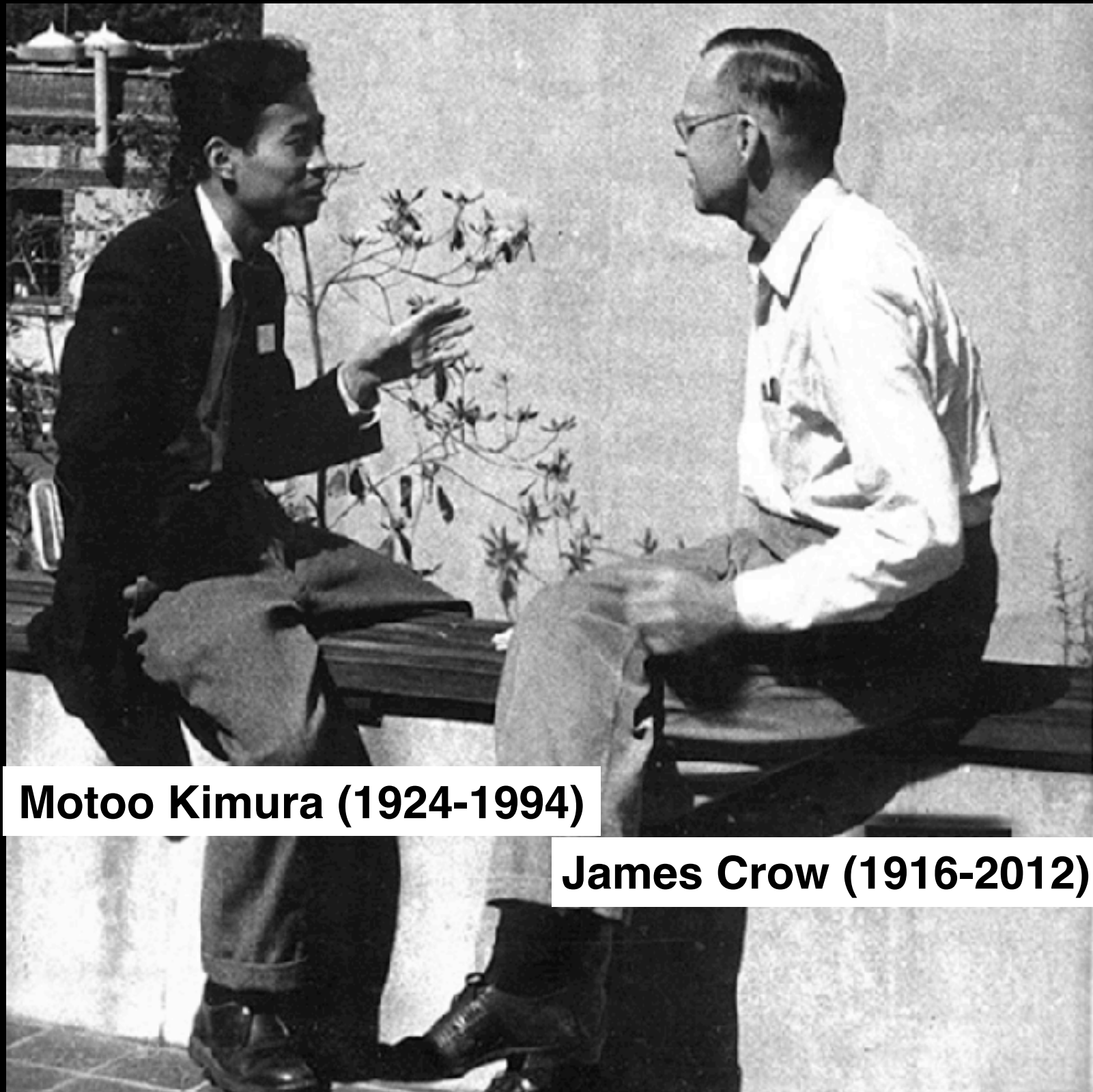
# Inheritance of variation

Adding selection



# How random is evolution?

4



**Motoo Kimura (1924-1994)**

**James Crow (1916-2012)**

# Does evolution play with dice?



**Bernard de Fontenelle (1657-1757)**

The universe is but a  
watch on a larger  
scale.

# Randomness in evolution

## Sampling error

The difference between the value found in a finite sample from a population and the true value in the population.

# Randomness in evolution

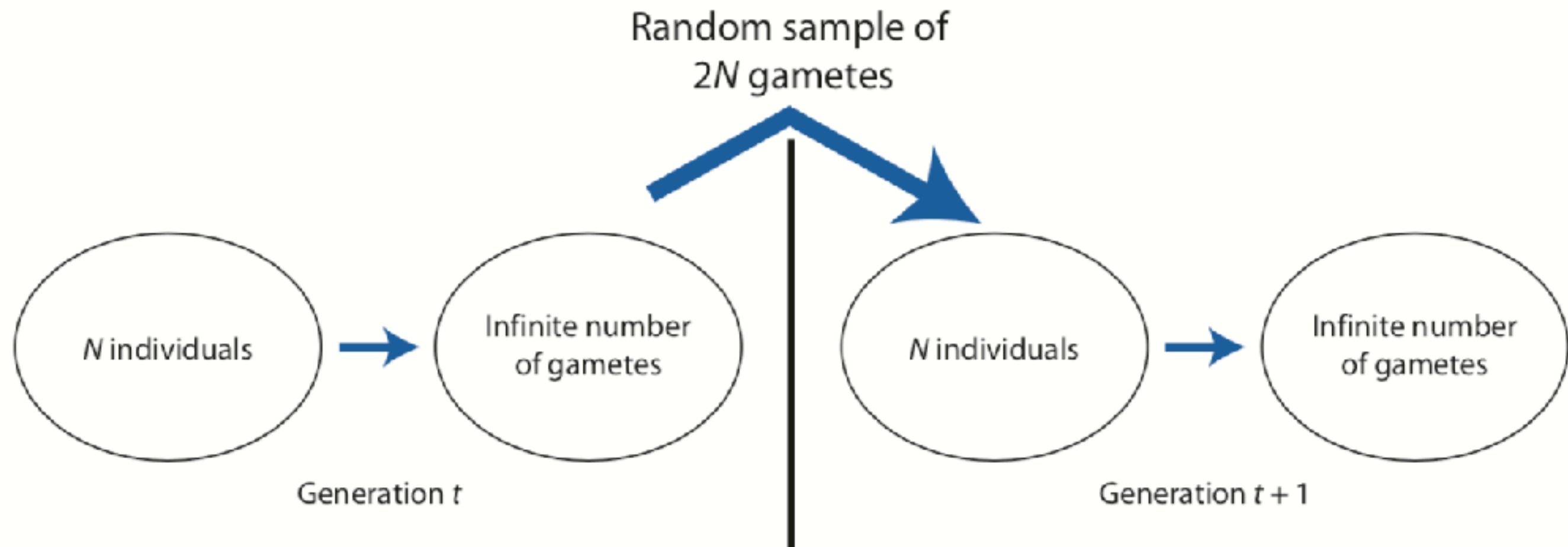
## Sampling error

The difference between the value found in a finite sample from a population and the true value in the population.

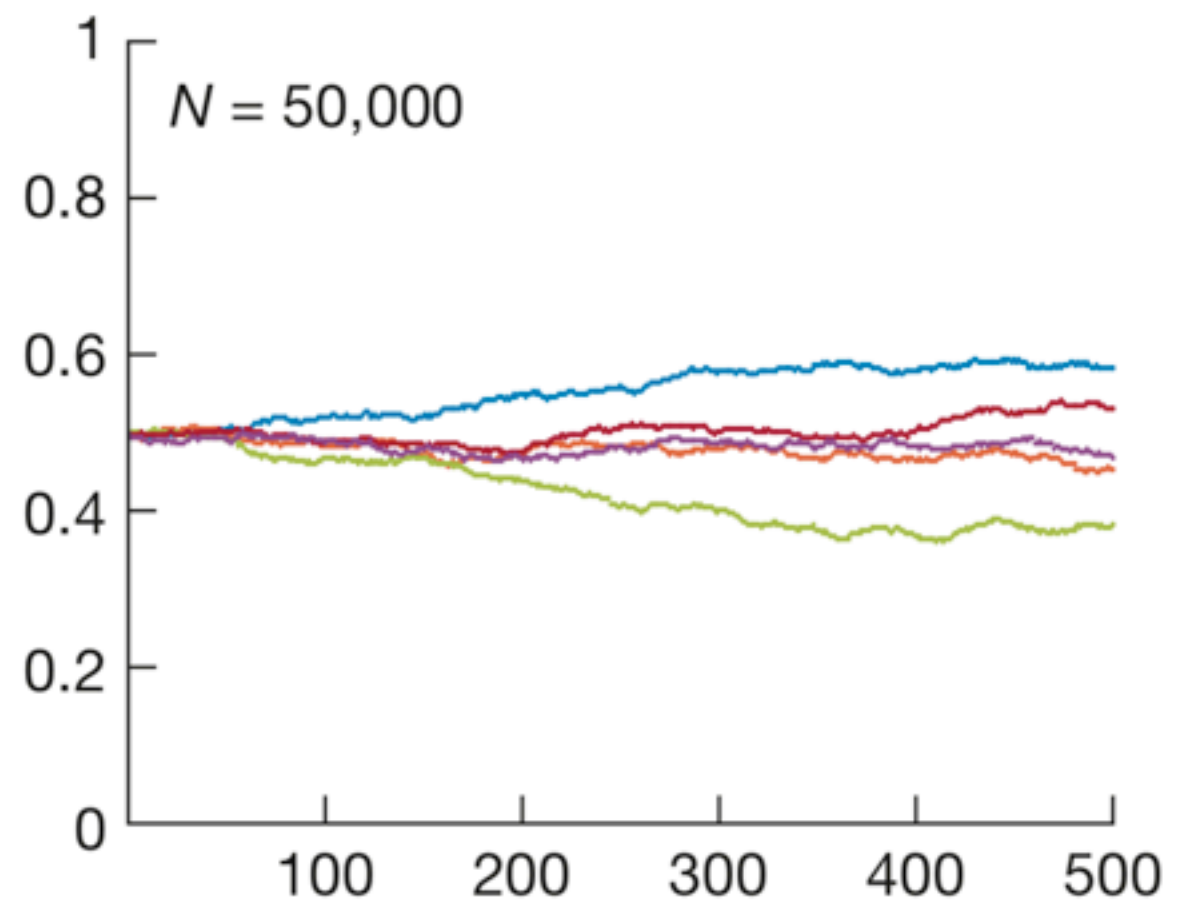
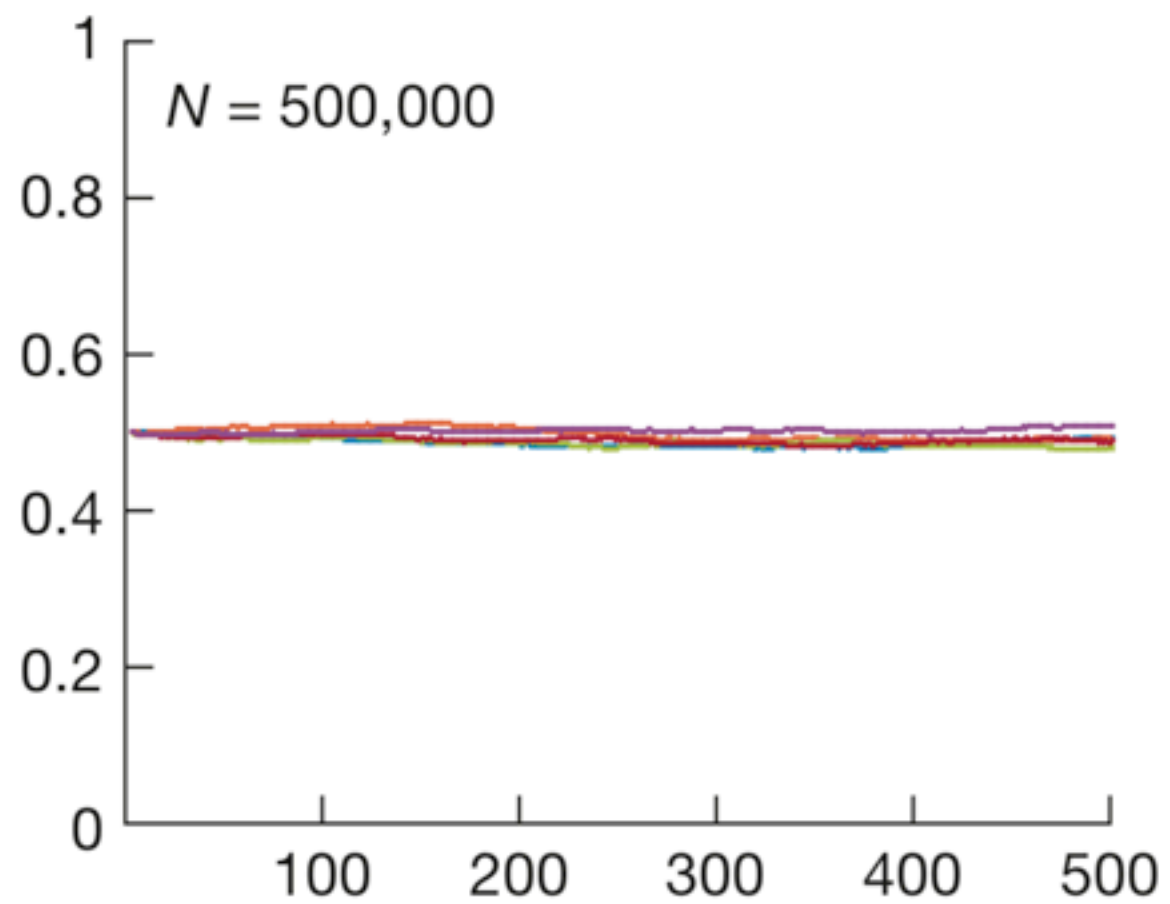
## Genetic drift

Random changes in allele frequency from one generation to the next in biological populations due to the finite samples of individuals, gametes, and ultimately alleles that contribute to the next generation.

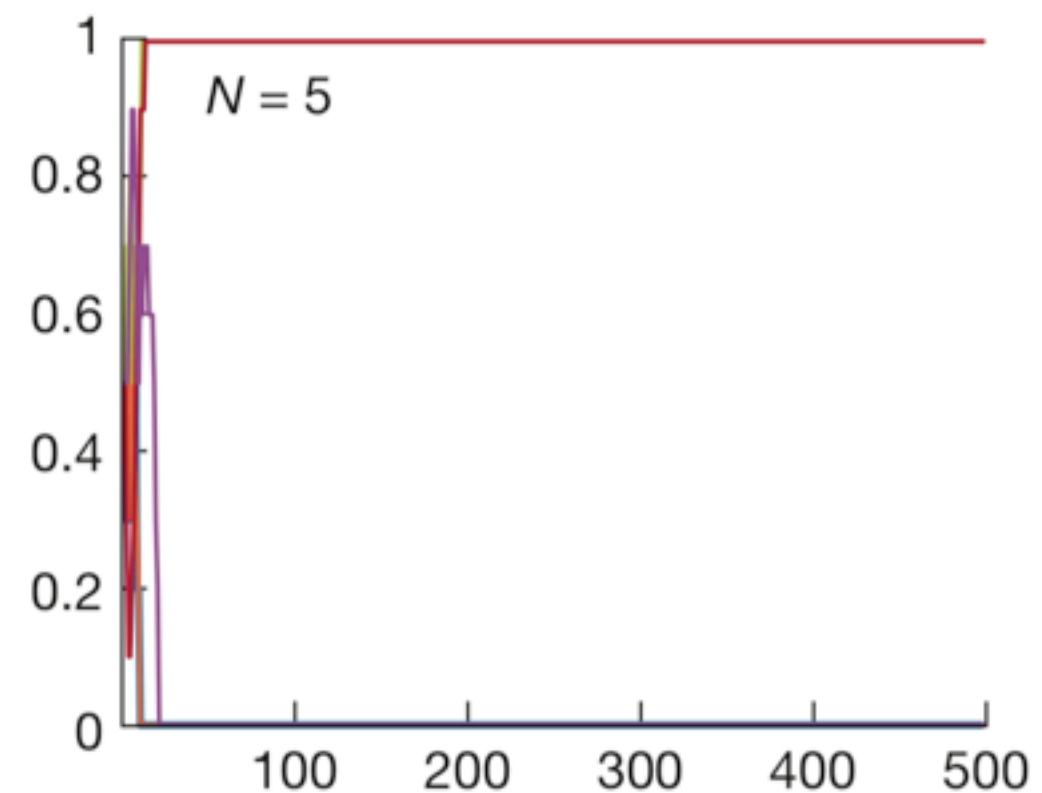
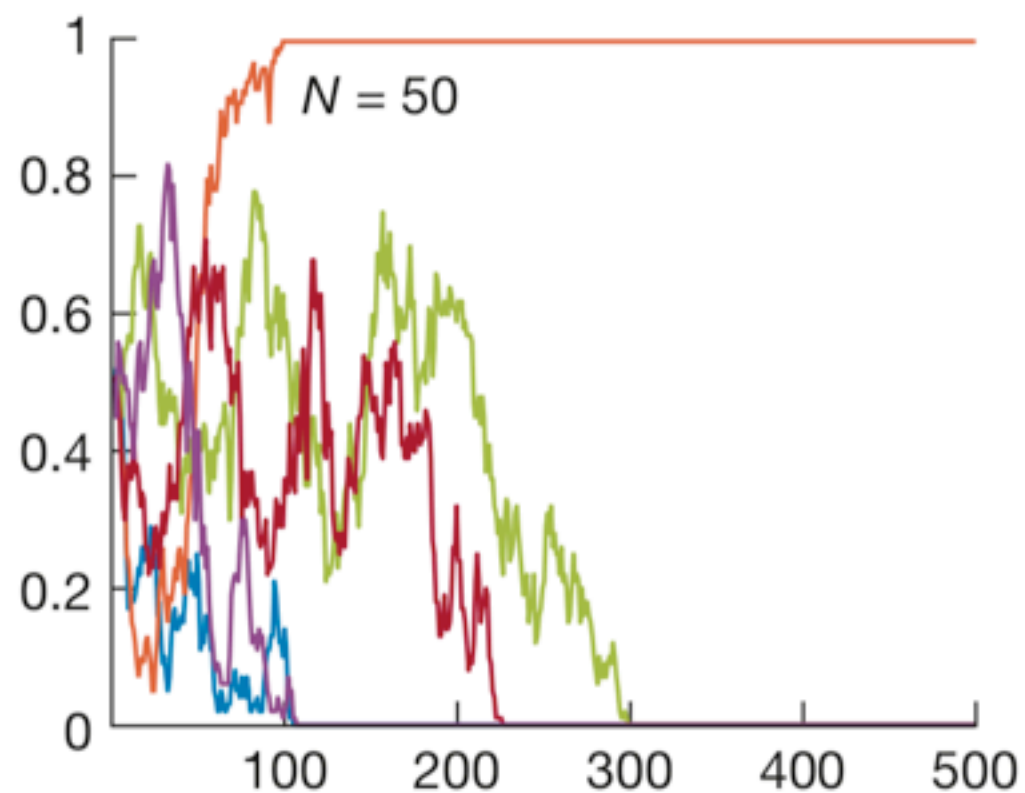
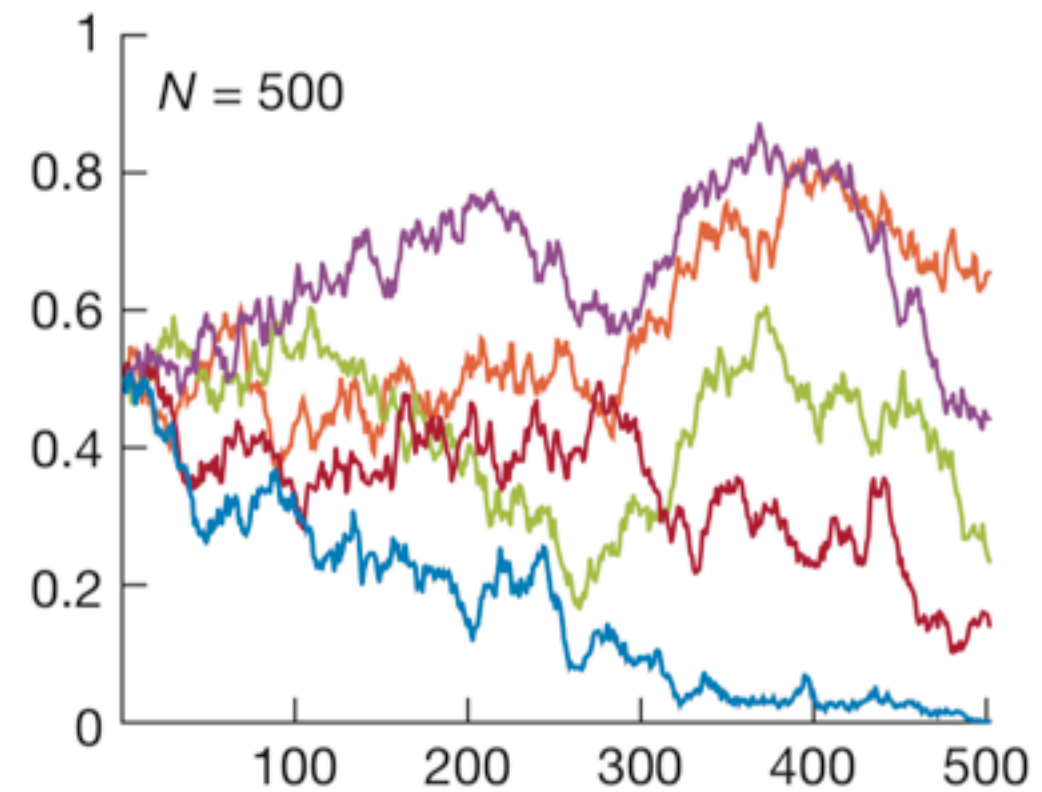
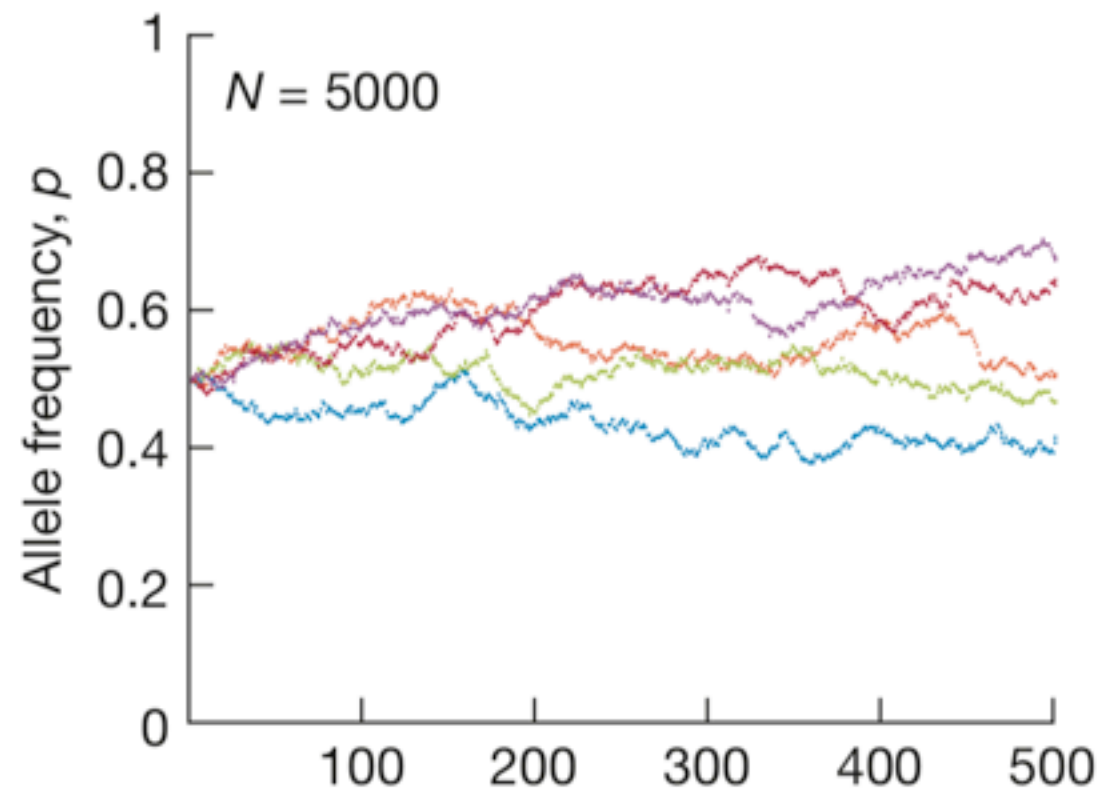
# Randomness in evolution



# Randomness in evolution



# Randomness in evolution



Generation

# Randomness in evolution

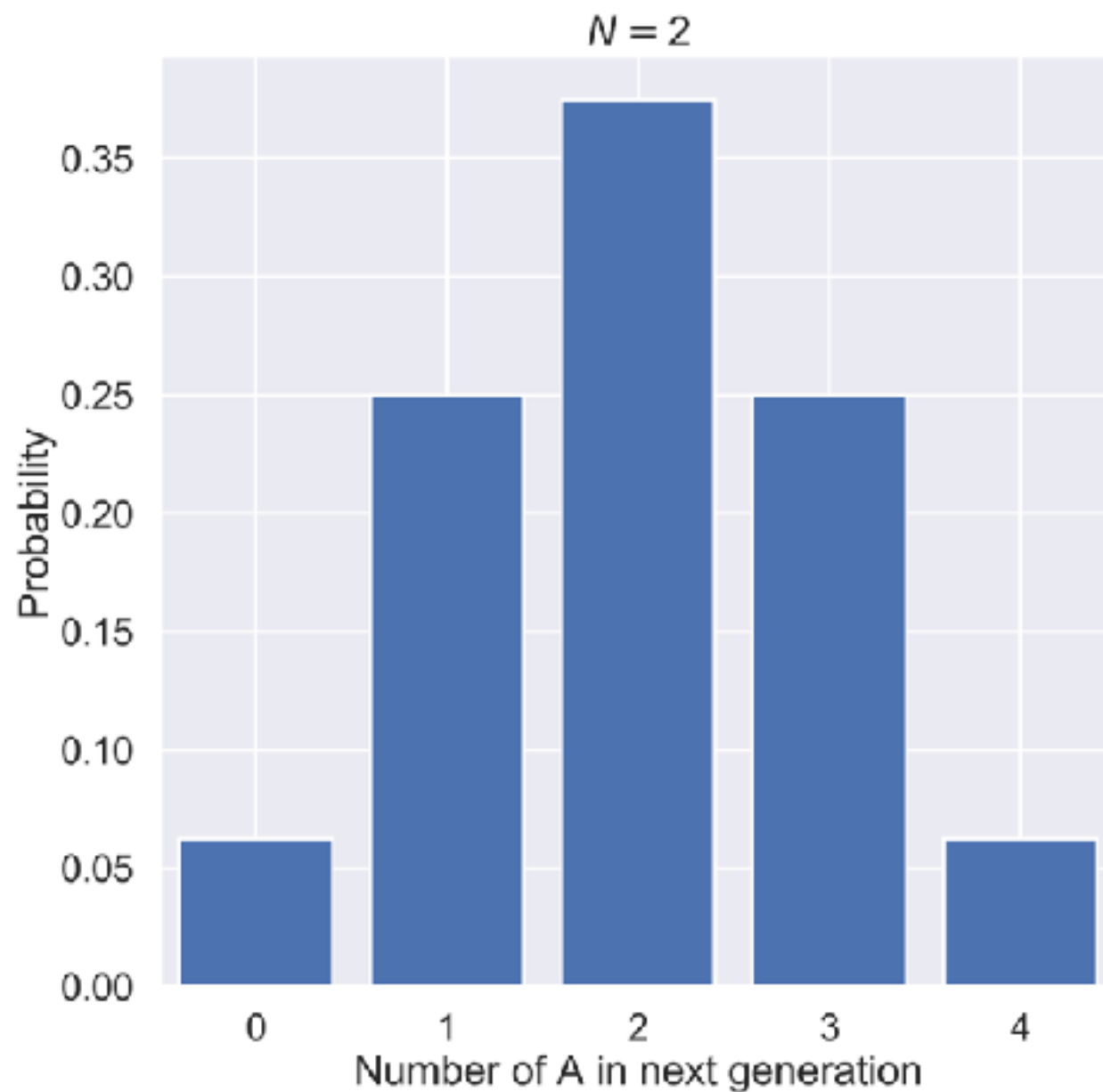
$$P_{(i=A)} = \binom{2N}{i} p^i q^{2N-i} \quad \text{where} \quad \binom{2N}{i} = \frac{(2N)!}{i!(2N-i)!}$$

## Bernoulli or binomial random variable

A variable representing a trial or sample that can have only two possible outcomes, such as zero or one.

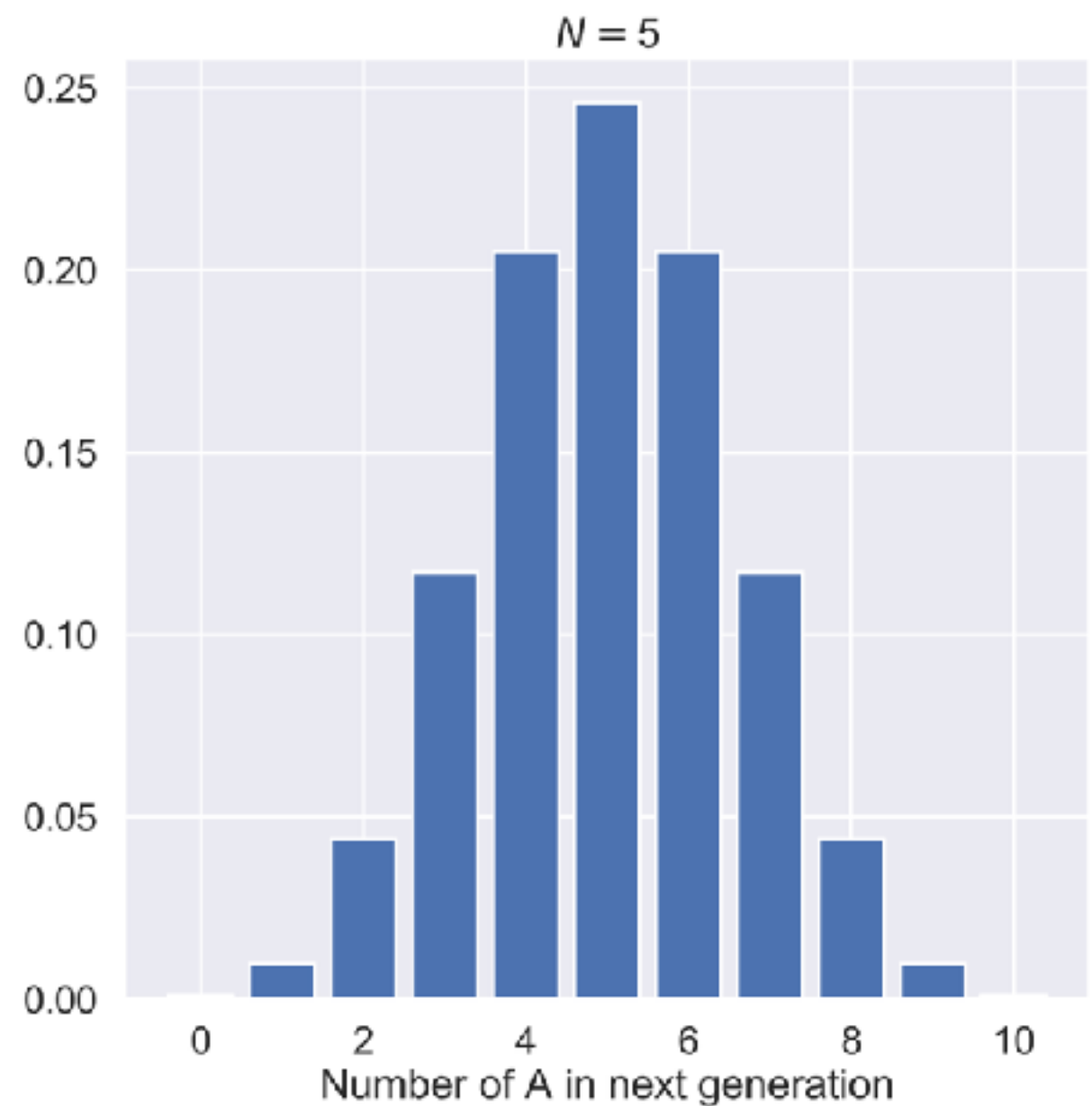
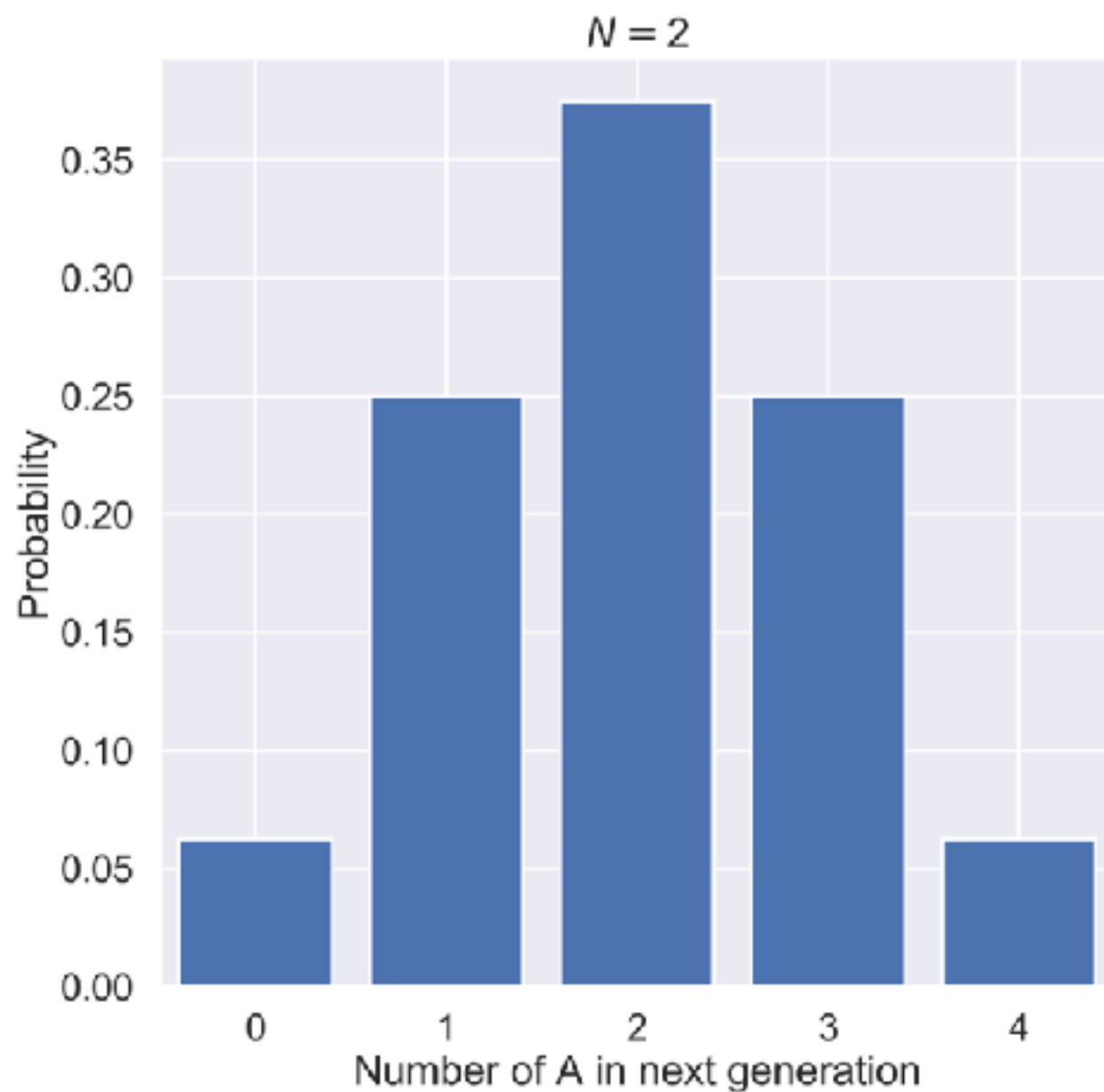
# Randomness in evolution

$$P_{(i=A)} = \binom{2N}{i} p^i q^{2N-i} \quad \text{where} \quad \binom{2N}{i} = \frac{(2N)!}{i!(2N-i)!}$$

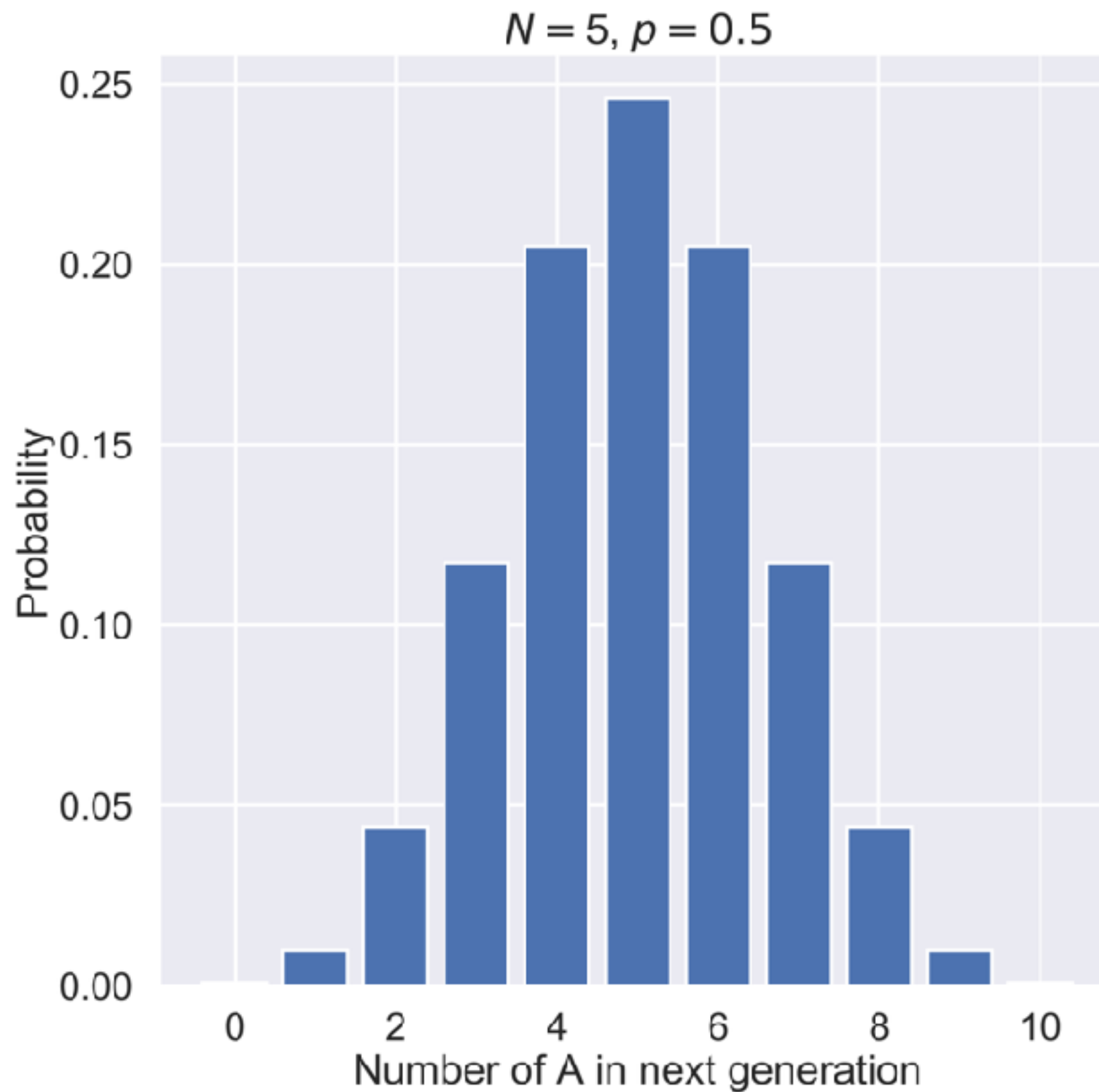


# Randomness in evolution

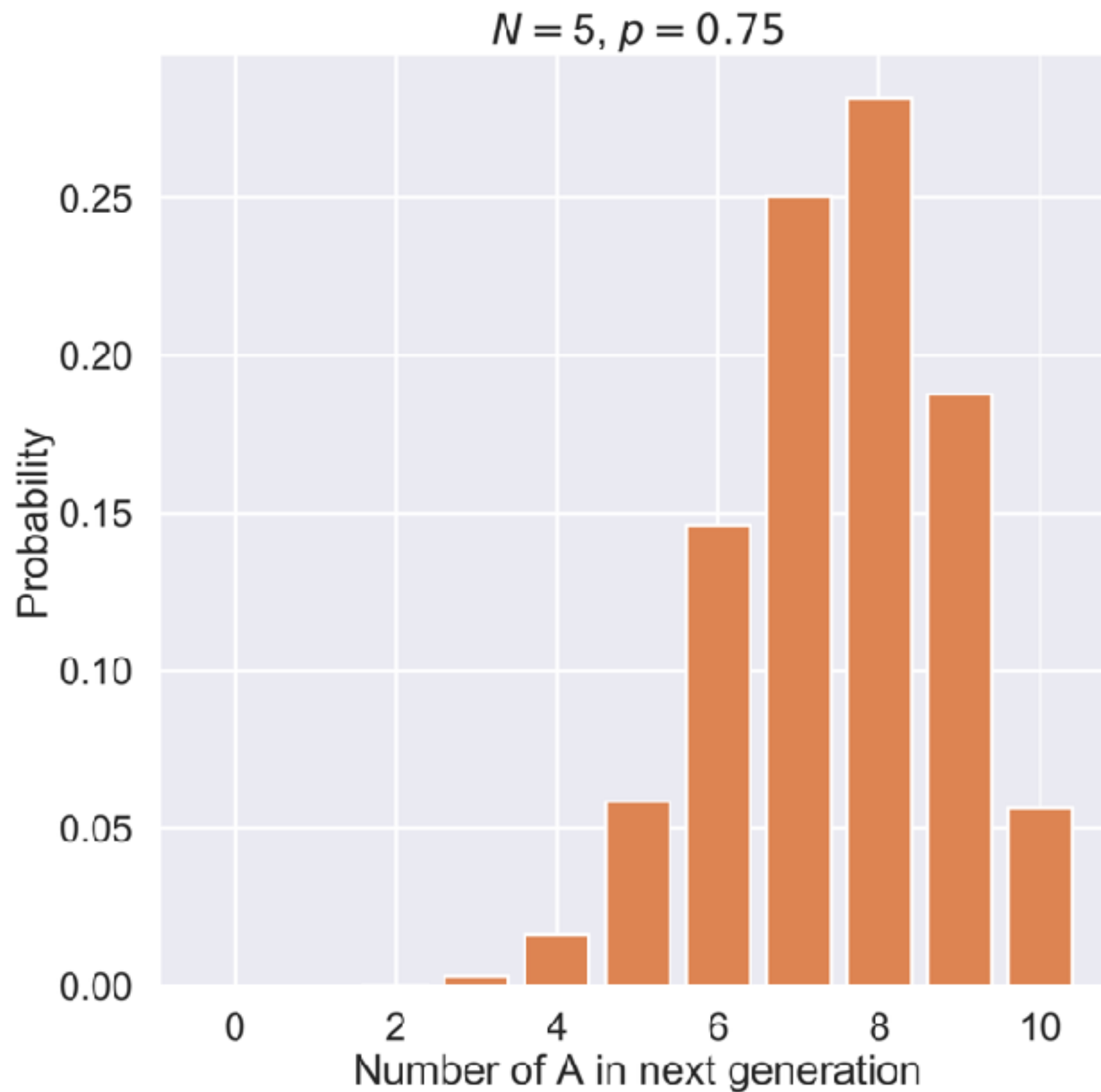
$$P_{(i=A)} = \binom{2N}{i} p^i q^{2N-i} \quad \text{where} \quad \binom{2N}{i} = \frac{(2N)!}{i!(2N-i)!}$$



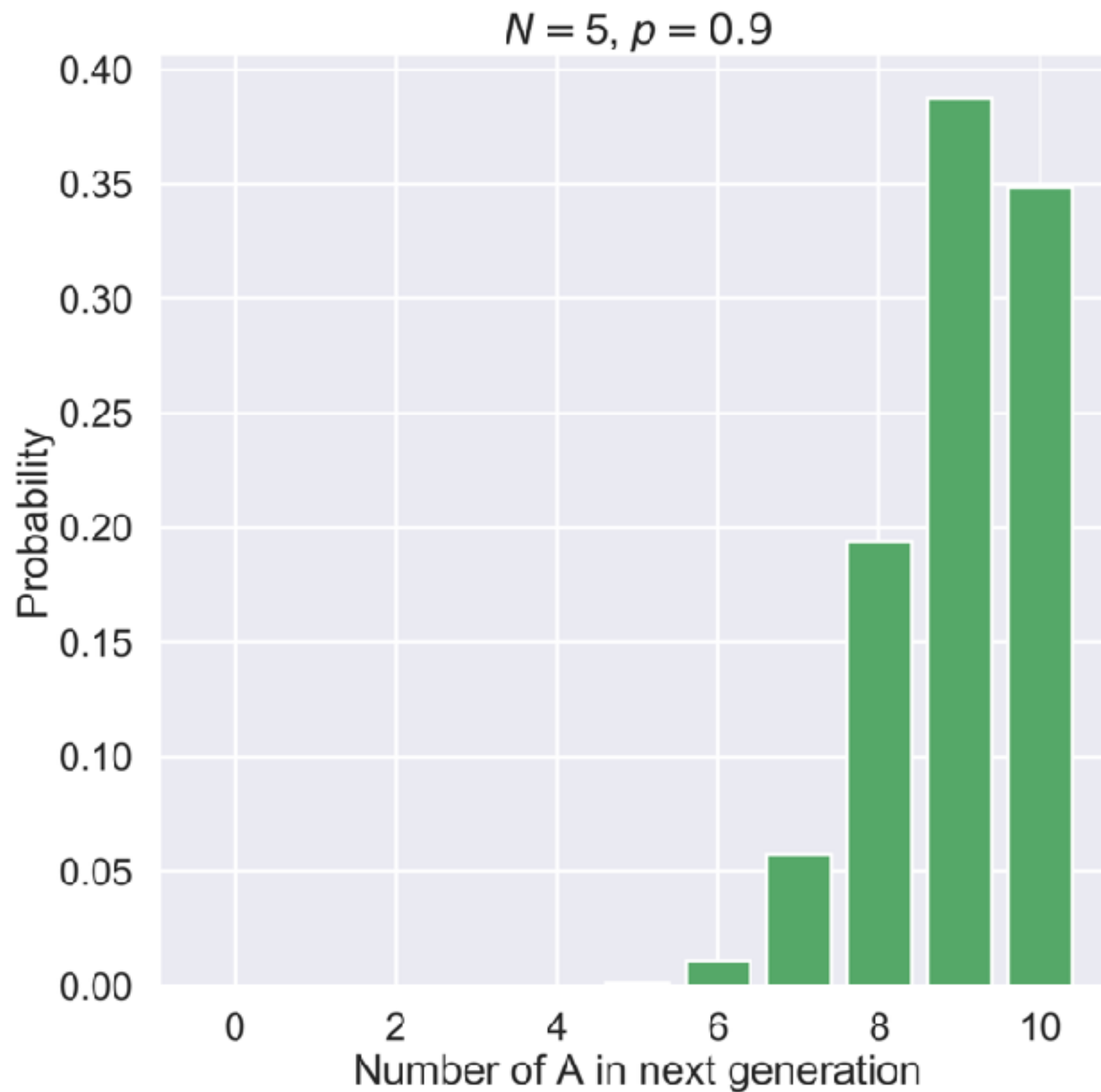
# Randomness in evolution



# Randomness in evolution



# Randomness in evolution



# Randomness in evolution

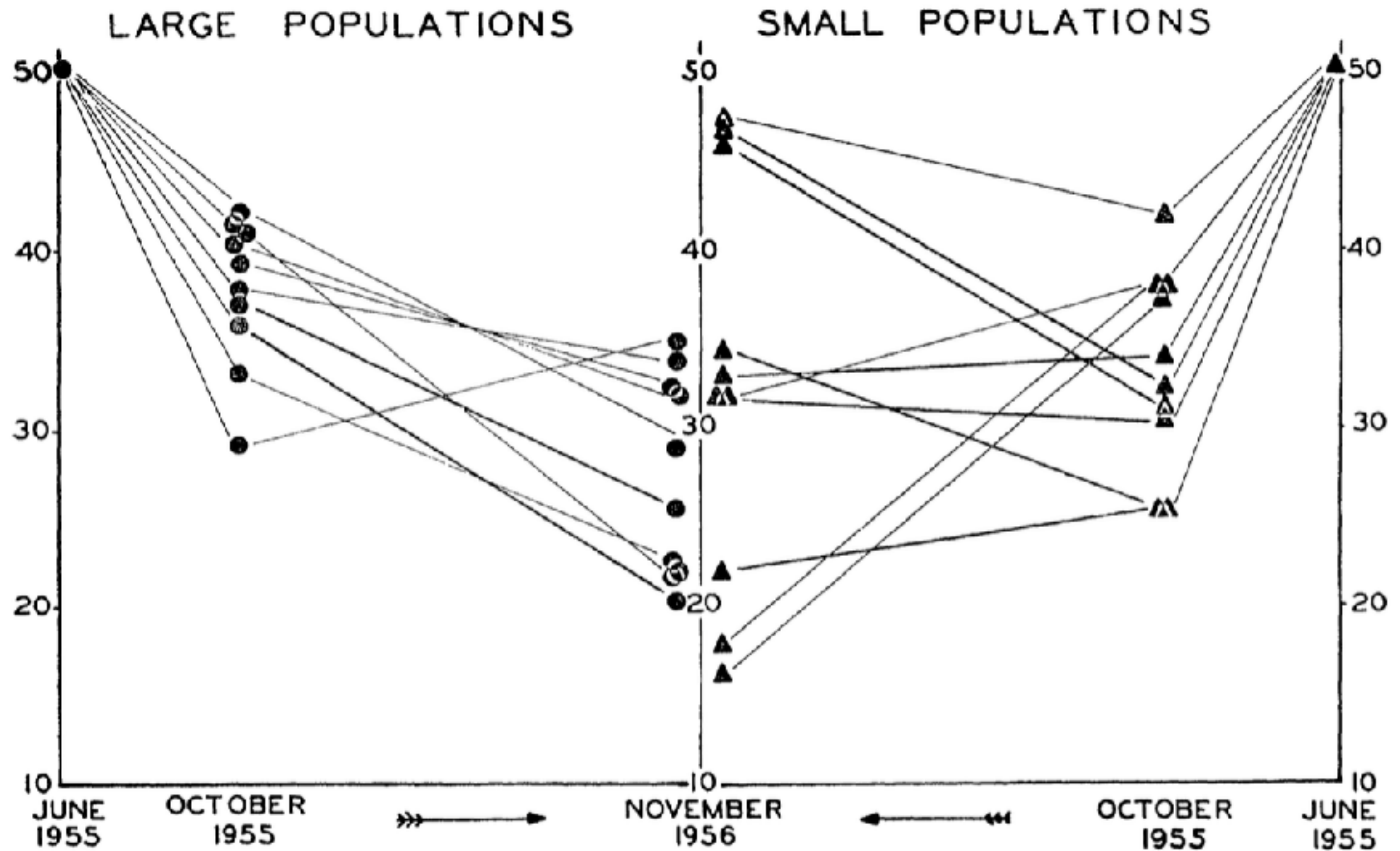
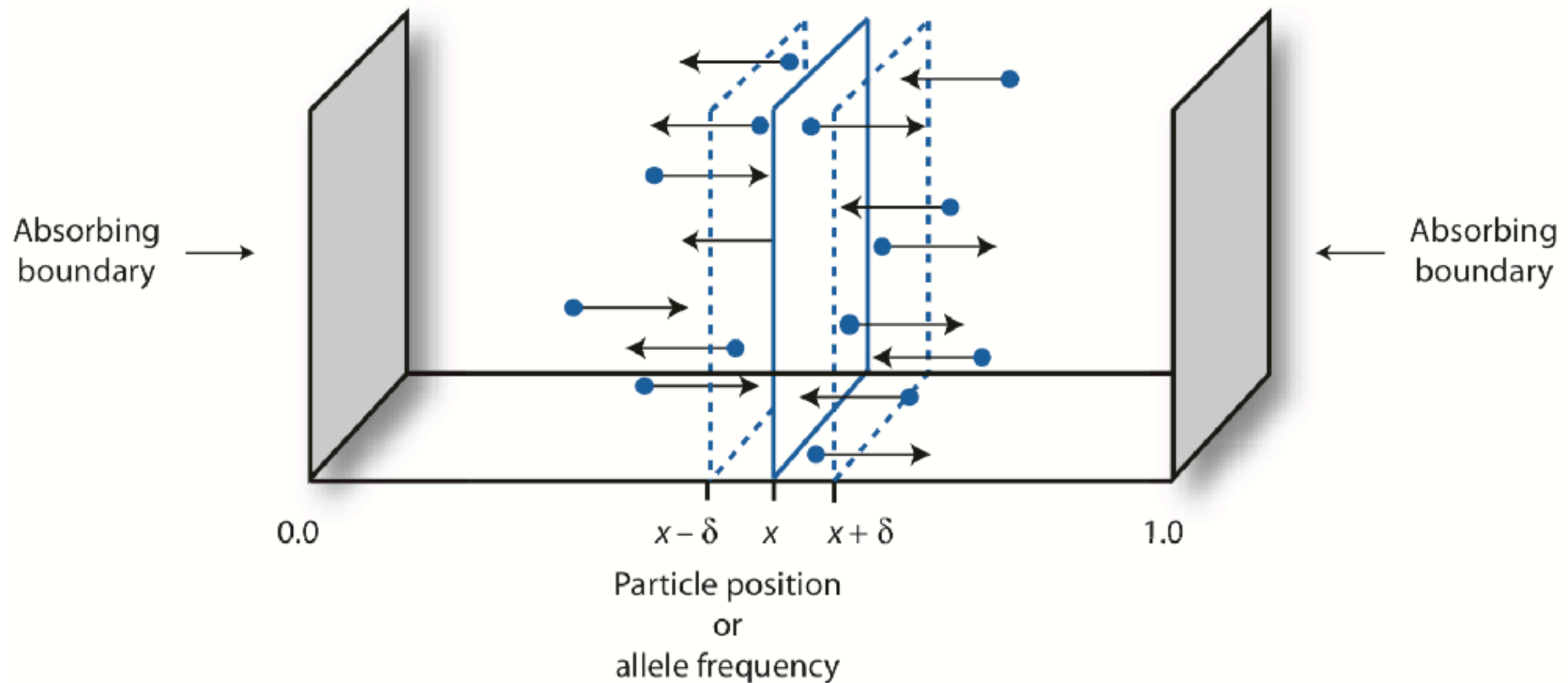


FIG. 2. The frequencies (in percent) of PP chromosomes in twenty replicate experimental populations of mixed geographic origin (Texas by California).

# Randomness in evolution

## Drift and natural Selection



# Randomness in evolution

## Drift and natural Selection

Probability of fixation for an allele

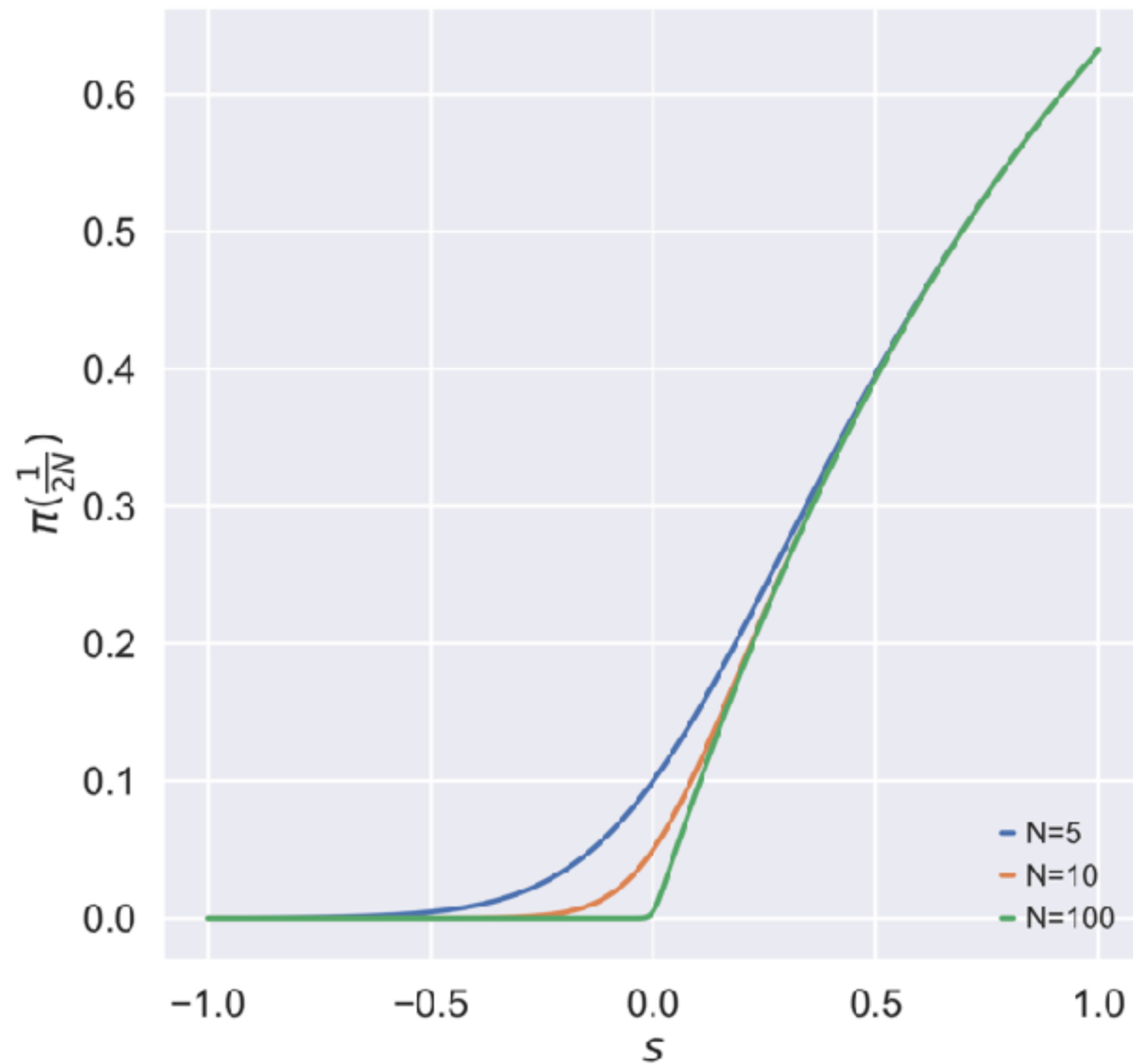
$$\pi(p) = \frac{1 - e^{-2Nsp}}{1 - e^{-2Ns}}$$

Probability of fixation for a new mutation

$$\pi\left(\frac{1}{2N}\right) = \frac{1 - e^{-s}}{1 - e^{-2Ns}}$$

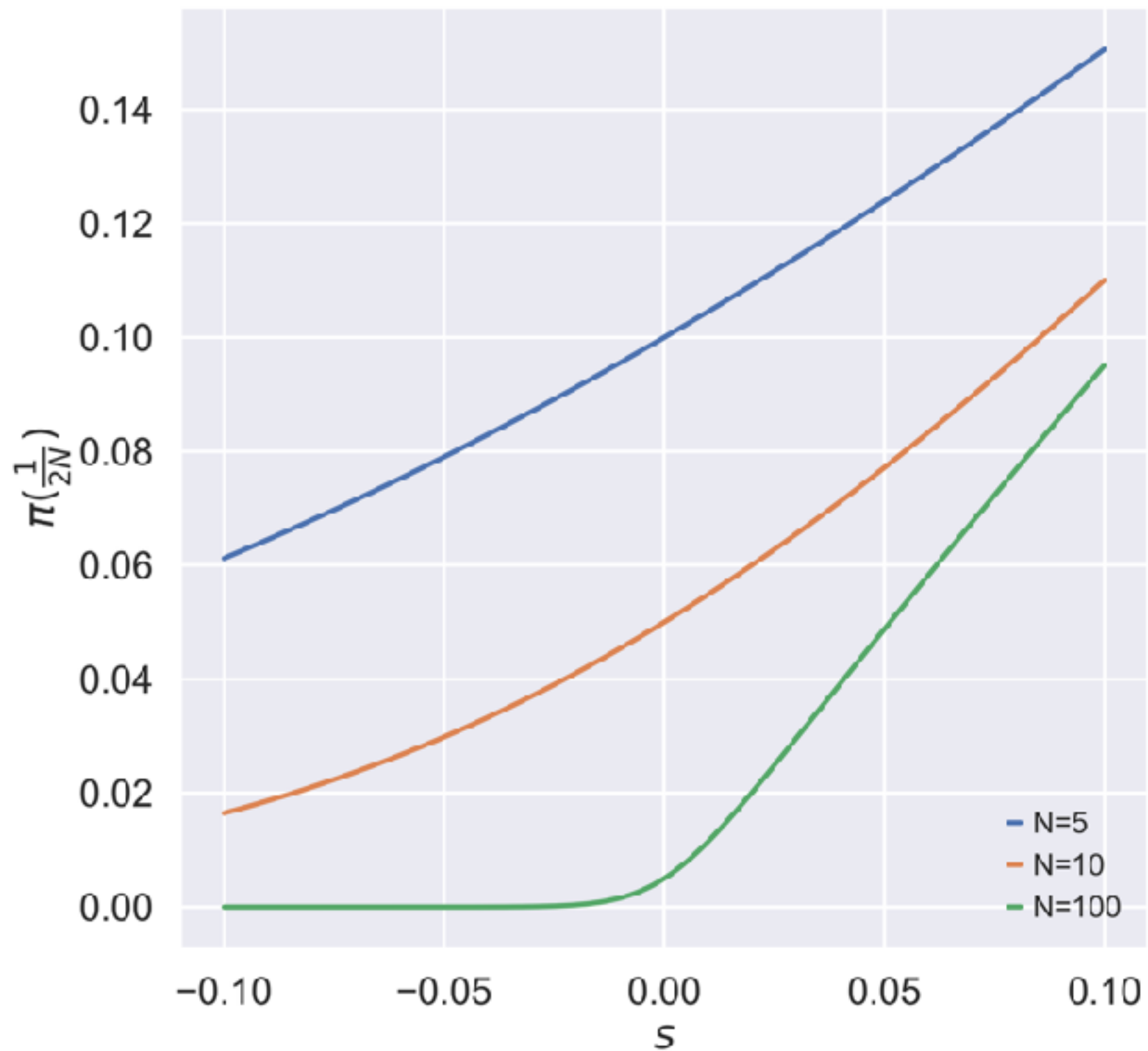
# Randomness in evolution

## Drift and natural Selection



# Randomness in evolution

## Drift and natural Selection



**Fin.**